DevOrSysAdminScripts

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

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Current git SHA1: 9c112177599a6d2582ed34deb49cc85abf49effd
Code lines: 9616 total lines, 8687 not empty lines, 929 empty lines.

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1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spell_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── DevOrSysAdminScripts.html.tpl
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_listing.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
```

```
│   ├── licenses_notes.txt
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 01234\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345
│   ├──
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│   │ 012345\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456\
│   ├──
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │ an_see_that_it_works.txt
│   ├── funny.sh
│   ├── spampty_data.txt
│   └──
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── README.html
```

```
├── README.md
├── README.md.tpl
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 73 files

[ 12K Mar  5 19:47]  ./
├── [4.0K Mar  5 19:47]  build_and_checks_dependencies/
│   ├── [3.8K Mar  2 18:25]  build_and_checks.exec.sh.tpl*
│   ├── [3.8K Mar  2 19:19]  build_md_from_printable_md.exec.sh*
│   ├── [2.8K Mar  3 08:41]  check_shell_scripts_beginnings.libr.sh
│   ├── [4.8K Mar  5 16:21]  check_shell_scripts_indentation.libr.sh
│   ├── [3.1K Mar  3 10:22]  check_URLs.libr.sh
│   ├── [ 29K Mar  5 19:45]  common_build_and_checks.exec.sh*
│   ├── [ 24K Mar  3 18:32]  common_build_and_checks.exec.sh.tpl
│   ├── [9.0K Mar  3 16:14]  common_options.libr.sh
│   ├── [4.8K Mar  3 19:09]  comparisons.libr.sh
│   ├── [ 18K Mar  5 19:44]  create_PDF.exec.sh*
│   ├── [3.6K Mar  5 19:33]  generate_from_template.libr.sh
│   ├── [1.9K Feb 19 17:18]  get_common_text_glob_patterns.libr.sh
│   ├── [2.4K Mar  3 10:22]  grammar_and_spell_check.libr.sh
│   ├── [4.0K Mar  3 19:13]  licenses_templates/
│   │   ├── [ 13K Mar  3 19:13]  build_licenses_templates.exec.sh*
│   │   ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│   │   ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│   │   └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│   ├── [4.8K Mar  3 18:49]  lines_counts.libr.sh
│   ├── [ 13K Mar  5 15:54]  lines_filters.libr.sh
│   ├── [2.7K Feb 19 16:16]  lines_maps.libr.sh
│   ├── [1.8K Feb 26 16:58]  overwrite_if_not_equal.libr.sh
│   ├── [2.3K Mar  3 10:30]  python_black_complement.libr.sh
│   ├── [2.6K Mar  3 10:22]  python_isort_complement.libr.sh
│   ├── [2.0K Mar  3 17:16]  repository_data.libr.sh
│   ├── [1.6K Feb  3 17:57]  split_score.exec.php
│   ├── [1.9K Feb  3 20:11]  split_score.libr.php
│   ├── [ 13K Mar  5 16:40]  strings_functions.libr.sh
│   ├── [3.4K Mar  4 17:55]  too_long_code_lines.libr.sh
│   ├── [4.2K Mar  3 17:18]  update_common_build_and_checks.exec.sh*
│   └── [5.5K Mar  5 16:43]  update_or_check_files_names_listing.exec.sh*
├── [4.0K Mar  5 19:45]  build_and_checks.exec.sh*
├── [4.0K Mar  5 19:47]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
```

```
│   ├── [ 16K Mar  2 12:57]  DevOrSysAdminScripts.html.tpl
│   ├── [ 31K Mar  5 19:47]  DevOrSysAdminScripts.tex
│   ├── [8.5K Mar  2 13:01]  DevOrSysAdminScripts.tex.tpl
│   ├── [ 14K Mar  5 17:00]  files_listing.tex.tpl
│   ├── [5.0K Mar  5 16:36]  files_names_listing.txt
│   ├── [  24 Jan 15 22:26]  .gitignore
│   ├── [4.9K Mar  2 12:48]  licenses_notes.txt
│   ├── [1001 Feb 26 15:37]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.4K Mar  3 20:47]  repository_data.txt
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 167 Feb 19 18:39]  shellcheck.ini
│   └── [3.1K Mar  3 18:56]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [377K Mar  5 19:47]  DevOrSysAdminScripts.html
├── [341K Mar  5 19:47]  DevOrSysAdminScripts.pdf
├── [4.0K Mar  3 20:51]  git_helpers/
│   ├── [3.8K Mar  3 20:51]  git_checks.libr.sh
│   ├── [1.8K Feb 27 20:20]  post-commit*
│   ├── [ 13K Mar  3 10:50]  post_commit_renaming_comment.libr.sh
│   └── [1.4K Feb 19 18:42]  pre-commit*
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [ 830 Dec 29 19:10]  LiensUtiles.txt
├── [4.0K Mar  5 16:23]  miscellaneous/
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123\
│   ├── [   0 Mar  4 10:03]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 01234\
│   ├── [   0 Mar  4 10:01]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345\
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456\
│   ├── [   0 Mar  4 09:13]
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │ an_see_that_it_works.txt
```

```
│   ├── [1.3K Mar  5 16:23]  funny.sh
│   ├── [  94 Mar  3 19:56]  spampty_data.txt
│   └── [   0 Mar  4 09:13]
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── [7.4K Mar  1 22:30]  README.html
├── [3.7K Mar  1 22:30]  README.md
├── [3.7K Mar  1 22:29]  README.md.tpl
├── [9.1K Feb 26 15:05]  unicode.libr.php
├── [1.4K Feb  3 14:19]  upgrade_build_and_checks.exec.sh*
└── [1.4K Feb 19 14:41]  wget_sha512.libr.sh

6 directories, 73 files
```

2 Listing of files

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
  basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpmd;

- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
`https://www.gnu.org/software/findutils/`,
`https://savannah.gnu.org/projects/findutils/`.

grep (GNU grep) can be found at the following URLs:
`https://www.gnu.org/software/grep/`,
`https://savannah.gnu.org/projects/grep/`.

sed (GNU sed) can be found at the following URLs:
`https://www.gnu.org/software/sed/`,
`https://savannah.gnu.org/projects/sed/`.

ShellCheck can be found at the following URLs:
`https://www.shellcheck.net/`,
`https://github.com/koalaman/shellcheck`.

GNU Wget can be found at the following URLs:
`https://www.gnu.org/software/wget/`,
`https://savannah.gnu.org/projects/wget/`.

more of util-linux can be found at the following URLs:
`https://www.kernel.org/pub/linux/utils/util-linux/`,
`https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/`,
`https://github.com/util-linux/util-linux` official?

pandoc can be found at the following URLs:
`https://pandoc.org/`,
`https://github.com/jgm/pandoc`.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,
https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

If you see something that I missed regarding my dependencies, please tell me/email me :).

2.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
```

```
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    {
      echo "temp/"
      echo ""
    } > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  @sha512_common_build_and_checks.exec.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}
```

```
if [[ "$*" == *--fixed_point_build* ]]; then
  echo "--fixed_point_build"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed_point_build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.2 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  local LFBFL_base_name="README"
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --base-name=* ]]; then
      LFBFL_base_name=${LFBFL_arg#--base-name=}
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        echo "Searching md file: ${LFBFL_base_name}"
      fi
      break
    fi
  done
  readonly LFBFL_base_name

  # Remove line returns here to keep lines short.
```

```
local LFBFL_sed_expression
LFBFL_sed_expression='s/(\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
LFBFL_sed_expression+=';s/(<http[^\n\\]*)\\\n/\1/Mg'
LFBFL_sed_expression+=';s/(- <http[^\n\\]*)\\\n/\1/Mg'
readonly LFBFL_sed_expression
# Since Markdown can contain code, it is limited to Markdown URLs.
# Hence it is way more complicated than 's/\\\n//Mg' expression
# in other files.
# Another downside is that more than one pass is needed.

if [[ -f "${LFBFL_base_name}.md.tpl" ]];
then
  sed -Ez -e "${LFBFL_sed_expression}"\
          -e "${LFBFL_sed_expression}"\
          -e "${LFBFL_sed_expression}"\
          -e "${LFBFL_sed_expression}"\
    "${LFBFL_base_name}.md.tpl"\
    > "${LFBFL_base_name}.md.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.md"\
    "${LFBFL_base_name}.md.temp"
else
  echo "No file ${LFBFL_base_name}.md.tpl"
fi

pandoc --from markdown --to html --standalone\
  --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
  -o "${LFBFL_base_name}.html.temp"
# Correcting pandoc HTML "by hand".
# https://github.com/jgm/pandoc/issues/10957
# With the most open-minded answer found on Internet ;) XD.
# This second sed expression is what I need,
# but it may be unsuitable for you.
# Other problems:
# https://github.com/jgm/pandoc/issues/11484
local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
readonly LFBFL_prewrap
sed -i -E -e 's~(  <meta .*) />~\1>~g'\
          -e 's~<html .*>~<html lang="en">~'\
          -e '/    code\{white-space: pre-wrap;\}/d'\
  "${LFBFL_base_name}.html.temp"
sed -i -Ez -e 's~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
           -e "${LFBFL_prewrap}"\
  "${LFBFL_base_name}.html.temp"
overwrite_if_not_equal "${LFBFL_base_name}.html"\
  "${LFBFL_base_name}.html.temp"
```

```

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

2.3 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # declare -r LFBFL_file_path=$(basename "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
```

```
  #   return 0
  # fi
  # diff <(head -n 1 "$1") <(echo '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  # diff <(head -n 1 "$1") <(echo '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head -n 1 "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || echo "$1:File has wrong shell script beginning."
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  # shopt -s globstar
  local LFBFL_file_path
  local LFBFL_head

  # for LFBFL_file_path in **/*.sh; do
  find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && echo "${LFBFL_file_path}:Checking shell script beginning."
    LFBFL_head=$(head -n 1 "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || echo "${LFBFL_file_path}:File has wrong shell script beginning."
  done
}
```

2.4 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_one_shell_script_indentation(){
  declare -i LFBFL_file_with_error=0
  local LFBFL_some_line
  local LFBFL_previous_line="workeduntilsomeasshole"
  # created a command named "workeduntilsomeasshole"
  # and called it without neither indentation, nor parameters
  # in a .sh script XD.
  # Manual justification of the lines L-4 and L-3 was a serendipity.
  # I just tried a joke and was lucky ;) :).
  local LFBFL_line_end
  local LFBFL_previous_line_end
  declare -i LFBFL_line_length
  declare -i LFBFL_line_end_length
  declare -i LFBFL_line_start_length
```

```
  local LFBFL_spacing
  declare -i LFBFL_offset
  local LFBFL_substring1
  local LFBFL_substring2
  declare -r LFBFL_pattern='^ *-e '
  # This grep would have been enough without some sed commands.
  declare -r LFBFL_s_lines=$(
    grep -EHn -B 1 '^(  )* ([^ ]|$)' "$1"
  )
  declare -a LFBFL_arr_lines
  mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
  readonly LFBFL_arr_lines

  for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
    if [[ "${LFBFL_some_line}" == "--" ]]; then
      # When switching between code fragments, we reinit.
      LFBFL_previous_line="workeduntilsomeasshole"
      continue
    fi
    # Note how it's funny that this loop always work because we check
    # shell scripts beginning in another script and thus we always
    # have a line before.
    if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]];
    then
      LFBFL_previous_line="${LFBFL_some_line}"
      continue
    fi
    # Remove the name of the file and then the line number
    LFBFL_line_end=${LFBFL_some_line#*:}
    LFBFL_line_end=${LFBFL_line_end#*:}
    # But for the previous line '-' or ':' is used instead of ':'
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_line_length=${#LFBFL_some_line}
    LFBFL_line_end_length=${#LFBFL_line_end}
    LFBFL_line_start_length=$((
      LFBFL_line_length-LFBFL_line_end_length
    ))
    LFBFL_offset=$((LFBFL_line_start_length-2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_offset+=1));;
      :) ((LFBFL_offset+=1));;
      [0-9]) ((LFBFL_offset+=2));;
```

```
      *) echo ':-(';;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        echo "${LFBFL_some_line}"
        LFBFL_file_with_error=1
      fi
    else
      echo "${LFBFL_some_line}"
      LFBFL_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_file_with_error -eq 1 ]]; then
    echo "$1:File has some lines with odd number of spaces."
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  # shopt -s globstar
  local LFBFL_file_path

  # for LFBFL_file_path in **/*.sh; do
  find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
```

```
    [[ LFBFL_i_verbose -eq 1 ]]\
      && echo "${LFBFL_file_path}:Checking shell script indentation."
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

2.5 build_and_checks_dependencies/check_URLs.libr.sh

```
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# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
```

```
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )

  local LFBFL_pattern
  local LFBFL_file_name
  local LFBFL_base_file_name
  local LFBFL_substitution
  local LFBFL_substitution2
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 0 ]]\
      || echo "Iterating on pattern: ${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H 'http:'\
      | grep -v "| xargs grep -H 'htt"\
      | grep -vP "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"
    # Last grep just above will remove false positives from
    # substitutions that fit on one line.
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find\
      | while read -r LFBFL_file_name;
    do
      [[ -f "${LFBFL_file_name}" ]] || continue
      LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      [[ LFBFL_i_verbose -eq 0 ]]\
        || echo "Handling the file: ${LFBFL_file_name}"
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=\
${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep -q "${LFBFL_substitution}" "${LFBFL_file_name}"; then
          sed -i "s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
```

```
            "${LFBFL_file_name}"
        fi
      done
    done
  done
}
```

2.6 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
```

```
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  # declare -r LFBFL_dependencies_URL="$2" too long
  declare -r LFBFL_start_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  declare -ar LFBFL_some_common_options=(
    "${LFBFL_verbose}"
    # "$--work-directory={LFBFL_work_directory}" doesn't work
    "${LFBFL_work_directory_option}"
  )

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  local LFBFL_file_name
  local LFBFL_URL
  local LFBFL_file_path

  LFBFL_file_name="build_md_from_printable_md.exec.sh"
  # LFBFL_script_download_URL
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='debb2bdda6f77ef82be5c17a547077d5c3db2bea62f92'
  LFBFL_correct_sha512+='6b092815c4d7a2dbd9b8461923bff8e38199d776a71b'
  LFBFL_correct_sha512+='c8d417fa396a556898c3d783681afb8c235922f'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='13346ffc2efcff3d0628b2a28855d1cd752f2ad3882fe'
  LFBFL_correct_sha512+='9aa3dca9c50c96e0b2780b5107d9bb7cf90eb885a80b'
  LFBFL_correct_sha512+='6ed84ea329f6d7c06ba0e8c368e78ed9ecfe2ac'
```

```
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='2ddb8563896f04c04d033965915082526b22ddf367356'
  LFBFL_correct_sha512+='dd10505286f459876e639cbd5f5a328ee0e39939f010'
  LFBFL_correct_sha512+='fac0833224c786d18abffd1f20c9ae84c8e7f9f'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="check_URLs.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='32ff1c210a5d6e6589ec7a3a692eca090759bf9692756'
  LFBFL_correct_sha512+='09dda93607349432d233674c1c79945ed92b79287c5e'
  LFBFL_correct_sha512+='d044ce240b872e35462edebb60cc9f27f218ba3'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="common_options.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='1c5b17cbbeb7d3018f4d1bf1ce2d4c7452ce3fa8557c2'
  LFBFL_correct_sha512+='a2391c448181b51e336c03fafdb1c51b9d7168837773'
  LFBFL_correct_sha512+='fe15a1235ada2337efae17e54e71e99ec313760'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="comparisons.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='445d448b0077ac8ff660bcdb8b5e3fb391a08dec1832d'
  LFBFL_correct_sha512+='1f6633807806d5b145aa6fda4c7835a6510988ce3c37'
  LFBFL_correct_sha512+='439b1fca19b27cb4757659cd1c204ed8aeb071f'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="create_PDF.exec.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='983acc9185fce5c064b475a234d0008c8c286fa4e9d6f'
  LFBFL_correct_sha512+='c8be96f49bb73f5e3a0d901f2487f2a704f4f481417a'
  LFBFL_correct_sha512+='bcc413088301404b50131964668978e2799a78b'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  LFBFL_file_name="generate_from_template.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='7d90727e83305c1269ed80ab59a138bb70fd36833bbb2'
  LFBFL_correct_sha512+='3713a9fcf8910fb1578c6f089c107f351a4c0fdad048'
  LFBFL_correct_sha512+='73c7d5870c45d4d5f1364af543d33eaecadd4f3'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='4e0915efcc9a4088b1f79c9c7efbf7da5d68e8e1b21ff'
  LFBFL_correct_sha512+='feb4746c9d468be80affa30babf35938e15f9161db3e'
  LFBFL_correct_sha512+='3f6d95c913b86638563d7bd2995dc58a26f8e3a'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="grammar_and_spell_check.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='6deeaae24bc1f67bfe93acbf477d8ce15311edbb6b664'
  LFBFL_correct_sha512+='68052a4e29e326c33977272f4f8a6ede39cd1065f012'
  LFBFL_correct_sha512+='c3171f253c230c59fc01c95f8a288eb90cb78ca'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  # /licenses_templates/ ---------------------------------------------
  declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
  declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
  LFBFL_file_name="build_licenses_templates.exec.sh"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='34c02434a90f63330b150ac55693b9a82baa9208a539c'
```

```
LFBFL_correct_sha512+='c6eec4cd48db32b28b32d42d6b089d148781dbaf4d49'
LFBFL_correct_sha512+='52b7b1ffd0efc8d1badbd90478d968725388cc0'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='809597a887cfd33d0556c3353ebd3d7cfac175cbd1d63'
LFBFL_correct_sha512+='c4181554009d81bf35b85ab068405b009f04c0a37107'
LFBFL_correct_sha512+='24db5af77178dbdbdfa18586cd5a61849f22821'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
```

```
  LFBFL_file_name="lines_filters.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='739357424121ecb80696cc5620ce4b51ec6e2b2d7c31e'
  LFBFL_correct_sha512+='968b6f8b45da6546042d405a2a725e7fe7e7f8467e1f'
  LFBFL_correct_sha512+='4904156d3e9d5b6947f739313fa9571599054fe'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="lines_maps.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='25c3e5082951260295b5f0e87ba1f2c978efdea1e8926'
  LFBFL_correct_sha512+='072ce4d2062dc2a47bb92b9098f081e553eeae5d9154'
  LFBFL_correct_sha512+='fdc50c9d88ae5365b776a23eb6e026b884a8374'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="overwrite_if_not_equal.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='2781a564d264bae84c787321481dc83ae43a1ce324ee6'
  LFBFL_correct_sha512+='d5ceb73d5a98607fad5fe9ff8d03979b898f7cd7c712'
  LFBFL_correct_sha512+='8215d26e76874d56aa67fb63fa9de776617d24a'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="python_black_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='ce29513479932cc7c2f8d79fc10ef2f3803b578c18106'
  LFBFL_correct_sha512+='9d5ba6462734acfe9907240f1c0b0fa1fecae672e722'
  LFBFL_correct_sha512+='e72c1bdf94077693cb86af1bbc34f18fb4400fa'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="python_isort_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='c913be48fdd63c3c518ce342860adfeb2698866548905'
```

```
LFBFL_correct_sha512+='6aa3e3299fc81f0c418e30d5d6ead46804e9b426a39f'
LFBFL_correct_sha512+='22fffcb1c89f7c2a380915b66acaa54dba6af1d'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="repository_data.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='236fe88b20c3e900e840651e489f84534036262ae048b'
LFBFL_correct_sha512+='fcfdd494c98cbdba17d634e4c842c5b37866bc0332ae'
LFBFL_correct_sha512+='e322e8fc140e7e8e0f1b2209cf839b6d56066a7'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.exec.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2f7ff107a43716109b1acbd7d89870d05577af00a981a'
LFBFL_correct_sha512+='05f162a054c96387fb99544f8225c333b1212f74326c'
LFBFL_correct_sha512+='f9ec6196e796f6257dfbba9347ac8c57b4f09db'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.libr.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='1413898a4fb4f31ca17993fd6110f7e1f32adf31c22c4'
LFBFL_correct_sha512+='7fa46a31d3c1511a3112a611e04fac19bb838d4476d9'
LFBFL_correct_sha512+='ecf77a28dd48d3d2732df7392c50e38262852d7'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='30db86474889d21a62976c8d69fed289d1f0d5e33dc4d'
LFBFL_correct_sha512+='57b7bca7098bc48ed06a60a6392b3f1f8e34a0903cc0'
LFBFL_correct_sha512+='96207fafc18fdfbda28409afc1b10e1d4fe874b'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="too_long_code_lines.libr.sh"
```

```
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='12172cd22a13672131b8e1fbd2894e3f6be70d23751cc'
LFBFL_correct_sha512+='7602fff28cc41e94882354e7f1b97bd8a08cd3819b91'
LFBFL_correct_sha512+='ab00bc7b502542313034c8d966821b8ed707313'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='a93461e1f83560cc9be0e8881173c3eaf7b00d1ee4397'
LFBFL_correct_sha512+='751c9dbb683327746e7111c382c51f2d52e98d4f7cfd'
LFBFL_correct_sha512+='6a8e66f2ce32ec5c8382348addc1460cd3ab3d9'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spell_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
```

```
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

enhanced_set_shell_option pipefail\
  && trap 'enhanced_unset_shell_option pipefail' RETURN

echo "Building license headers"
"./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
  "${LFBFL_some_common_options[@]}"

echo "Building README.md"
"./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
  "--work-directory=${LFBFL_work_directory}"\
  "--base-name=README"\
  "${LFBFL_verbose}"

echo "Building other MarkDown files"
local LFBFL_some_directory
declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
find "${LFBFL_work_directory}" -name "*.md.tpl"\
  | relevant_find\
  | while read -r LFBFL_file_path;
do
  [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
  echo "Found template ${LFBFL_file_path}"
  LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
  LFBFL_file_name=$(basename "${LFBFL_file_path}")
  LFBFL_file_name=${LFBFL_file_name%.md.tpl}
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_some_directory}"\
    "--base-name=${LFBFL_file_name}"\
    "${LFBFL_verbose}"
done

declare -i LFBFL_will_popd
```

```
  pushd_to_work_directory
  LFBFL_will_popd=$?
  can_continue_after_enhanced_pushd || return

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  declare -i LFBFL_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_upgrade_venvs_ts
  declare -i LFBFL_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      LFBFL_upgrade_venvs=1
    fi
  else
    LFBFL_upgrade_venvs=1
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_upgrade_venvs=0
      fi
```

```
    fi
  fi

  echo "Running shellcheck"
  find . -name "*.sh"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
      "${LFBFL_file_path}"
  done

  echo "---Python---"
  echo "Running isort"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  echo "Running python_isort_complement"
  python_isort_complement "${LFBFL_some_common_options[@]}"

  echo "Running black"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

```

```
  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  echo "Running python_black_complement"
  python_black_complement "${LFBFL_some_common_options[@]}"

  echo "Probing if mypy should be runned"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_no_toml=1
  find . -name "pyproject.toml"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
```

```
      echo "Running mypy"
      LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
      mypy "${LFBFL_directory_path}"
    fi
    LFBFL_no_toml=0
  done
  if [[ LFBFL_no_toml -eq 1 ]]; then
    echo "Running mypy"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  echo "Running bandit"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini build_and_checks_variables/bandit.ini\
    -b build_and_checks_variables/bandit_baseline.json\
    -r .
  # Saving new baseline in temp if necessary.
  bandit --ini build_and_checks_variables/bandit.ini\
    -f json -o build_and_checks_variables/temp/bandit_baseline.json\
    -r .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  echo "Running pylint"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
```

```
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile build_and_checks_variables/pylintrc.toml\
    --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  echo "Running ruff"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config build_and_checks_variables/ruff.toml
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  echo "---Python end---"

  echo "---PHP---"
  echo "Running PHPMD"
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  local LFBFL_phpmd_baseline="build_and_checks_variables/"
```

```
LFBFL_phpmd_baseline+="phpmd_baseline.xml"
readonly LFBFL_phpmd_baseline

local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
readonly LFBFL_temp_phpmd_baseline

local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
LFBFL_phpmd_rulesets+="design,naming,unusedcode"
readonly LFBFL_phpmd_rulesets

phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
# Saving new baseline in temp if necessary.
rm "${LFBFL_temp_phpmd_baseline}"
phpmd --color --generate-baseline\
  --baseline-file "${LFBFL_temp_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
echo "---PHP end---"

echo "---JS---"
LFBFL_npm_lint_directories=""
grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
  --result-variable-prefix="LFBFL_"
if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
  echo "Running ESLint"
  local LFBFL_JS_directory
  echo "${LFBFL_npm_lint_directories}"\
    | sed -e 's/\\n/\n/g'\
    | while read -r LFBFL_JS_directory;
  do
    (cd "${LFBFL_JS_directory}" && npm run lint)
  done
fi
echo "---JS end---"

echo "Analyzing too long lines"
# shellcheck disable=SC2248
too_long_code_lines "${LFBFL_some_common_options[@]}" \
  --max-line-length=${LFBFL_max_line_length}

echo "Analyzing shell scripts beginnings"
check_shell_scripts_beginnings "${LFBFL_some_common_options[@]}" \
  | relevant_grep

echo "Analyzing shell scripts indentation"
```

```
  check_shell_scripts_indentation "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  echo "Analyzing URLs"
  check_URLs "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  echo "Analyzing strange characters: hover over in doubt"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  [[ LFBFL_will_popd -eq 0 ]] && popd_from_work_directory

  echo "Checking listed files"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  echo "Creating the PDF file of the listing of the source code"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  echo "Running Nu W3C HTML CSS and SVG validator"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
```

```
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_vnu_jar_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      echo $'/!\\ATTENTION: Nu W3C validator is too old./!\\'
      echo "Go get the latest version here:"
      echo "https://github.com/validator/validator/releases"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    declare -a LFBFL_files_for_Nu2
    mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
    readonly LFBFL_files_for_Nu2
    java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
      --verbose "${LFBFL_files_for_Nu2[@]}"
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.7 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# that it will be useful,
```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  # declare -r LFBFL_dependencies_URL="$2" too long
  declare -r LFBFL_start_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  declare -ar LFBFL_some_common_options=(
    "${LFBFL_verbose}"
    # "$--work-directory={LFBFL_work_directory}" doesn't work
    "${LFBFL_work_directory_option}"
  )

  source ./wget_sha512.libr.sh
```

```
59
60    LFBFL_subdir="build_and_checks_dependencies"
61    local LFBFL_file_name
62    local LFBFL_URL
63    local LFBFL_file_path
64
65    LFBFL_file_name="build_md_from_printable_md.exec.sh"
66    # LFBFL_script_download_URL
67    LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
68    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
69    @sha512_build_md_from_printable_md.exec.sh@
70    wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
71      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
72    chmod +x "./${LFBFL_file_path}"
73
74    LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
75    LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
76    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
77    @sha512_check_shell_scripts_beginnings.libr.sh@
78    wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
79      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
80
81    LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
82    LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
83    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
84    @sha512_check_shell_scripts_indentation.libr.sh@
85    wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
86      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
87
88    LFBFL_file_name="check_URLs.libr.sh"
89    LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
90    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
91    @sha512_check_URLs.libr.sh@
92    wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
93      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
94
95    LFBFL_file_name="common_options.libr.sh"
96    LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
97    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
98    @sha512_common_options.libr.sh@
99    wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
100     "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
101
102   LFBFL_file_name="comparisons.libr.sh"
103   LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
104   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
@sha512_comparisons.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_create_PDF.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_generate_from_template.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_get_common_text_glob_patterns.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="grammar_and_spell_check.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_grammar_and_spell_check.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

# /licenses_templates/ ---------------------------------------------
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_build_licenses_templates.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_AGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_GPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_LGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_counts.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_filters.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_maps.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_overwrite_if_not_equal.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
```

```
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="python_black_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_python_black_complement.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="python_isort_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_python_isort_complement.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="repository_data.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_repository_data.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="split_score.exec.php"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_split_score.exec.php@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="split_score.libr.php"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_split_score.libr.php@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="strings_functions.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_strings_functions.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="too_long_code_lines.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_too_long_code_lines.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_update_or_check_files_names_listing.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spell_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"
```

```
  enhanced_set_bash_option extglob
  # shellcheck source=repository_data.libr.sh
  source "./${LFBFL_subdir}/repository_data.libr.sh"
  # shellcheck disable=SC2154,SC2309
  [[ enhanced_set_bash_option_extglob_result -eq 0 ]]\
    && enhanced_unset_bash_option extglob

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  echo "Building license headers"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  echo "Building README.md"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_work_directory}"\
    "--base-name=README"\
    "${LFBFL_verbose}"

  echo "Building other MarkDown files"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  find "${LFBFL_work_directory}" -name "*.md.tpl"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    echo "Found template ${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
    LFBFL_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_file_name=${LFBFL_file_name%.md.tpl}
    "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
      "--work-directory=${LFBFL_some_directory}"\
      "--base-name=${LFBFL_file_name}"\
      "${LFBFL_verbose}"
  done

  declare -i LFBFL_will_popd
  pushd_to_work_directory
  LFBFL_will_popd=$?
  can_continue_after_enhanced_pushd || return

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

```

```
  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  declare -i LFBFL_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_upgrade_venvs_ts
  declare -i LFBFL_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      LFBFL_upgrade_venvs=1
    fi
  else
    LFBFL_upgrade_venvs=1
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_upgrade_venvs=0
      fi
    fi
  fi

  echo "Running shellcheck"
  find . -name "*.sh"\
    | relevant_find\
    | while read -r LFBFL_file_path;
```

```
  do
    shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
      "${LFBFL_file_path}"
  done

  echo "---Python---"
  echo "Running isort"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  echo "Running python_isort_complement"
  python_isort_complement "${LFBFL_some_common_options[@]}"

  echo "Running black"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
```

```
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  echo "Running python_black_complement"
  python_black_complement "${LFBFL_some_common_options[@]}"

  echo "Probing if mypy should be runned"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_no_toml=1
  find . -name "pyproject.toml"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
      echo "Running mypy"
      LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
      mypy "${LFBFL_directory_path}"
    fi
    LFBFL_no_toml=0
  done
  if [[ LFBFL_no_toml -eq 1 ]]; then
```

```
    echo "Running mypy"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  echo "Running bandit"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini build_and_checks_variables/bandit.ini\
    -b build_and_checks_variables/bandit_baseline.json\
    -r .
  # Saving new baseline in temp if necessary.
  bandit --ini build_and_checks_variables/bandit.ini\
    -f json -o build_and_checks_variables/temp/bandit_baseline.json\
    -r .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  echo "Running pylint"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_pylint_venv}/bin/activate"
```

```
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile build_and_checks_variables/pylintrc.toml\
    --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  echo "Running ruff"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config build_and_checks_variables/ruff.toml
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  echo "---Python end---"

  echo "---PHP---"
  echo "Running PHPMD"
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  local LFBFL_phpmd_baseline="build_and_checks_variables/"
  LFBFL_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_phpmd_baseline

  local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

```

```
565     local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
566     LFBFL_phpmd_rulesets+="design,naming,unusedcode"
567     readonly LFBFL_phpmd_rulesets
568
569     phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
570       . text "${LFBFL_phpmd_rulesets}"
571     # Saving new baseline in temp if necessary.
572     rm "${LFBFL_temp_phpmd_baseline}"
573     phpmd --color --generate-baseline\
574       --baseline-file "${LFBFL_temp_phpmd_baseline}"\
575       . text "${LFBFL_phpmd_rulesets}"
576     echo "---PHP end---"
577
578     echo "---JS---"
579     LFBFL_npm_lint_directories=""
580     grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
581       --result-variable-prefix="LFBFL_"
582     if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
583       echo "Running ESLint"
584       local LFBFL_JS_directory
585       echo "${LFBFL_npm_lint_directories}"\
586         | sed -e 's/\\n/\n/g'\
587         | while read -r LFBFL_JS_directory;
588       do
589         (cd "${LFBFL_JS_directory}" && npm run lint)
590       done
591     fi
592     echo "---JS end---"
593
594     echo "Analyzing too long lines"
595     # shellcheck disable=SC2248
596     too_long_code_lines "${LFBFL_some_common_options[@]}" \
597       --max-line-length=${LFBFL_max_line_length}
598
599     echo "Analyzing shell scripts beginnings"
600     check_shell_scripts_beginnings "${LFBFL_some_common_options[@]}" \
601       | relevant_grep
602
603     echo "Analyzing shell scripts indentation"
604     check_shell_scripts_indentation "${LFBFL_some_common_options[@]}" \
605       | relevant_grep
606
607     echo "Analyzing URLs"
608     check_URLs "${LFBFL_some_common_options[@]}" \
609       | relevant_grep
610
```

```
  echo "Analyzing strange characters: hover over in doubt"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  [[ LFBFL_will_popd -eq 0 ]] && popd_from_work_directory

  echo "Checking listed files"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  echo "Creating the PDF file of the listing of the source code"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  echo "Running Nu W3C HTML CSS and SVG validator"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_vnu_jar_ts))
```

```
    ((LFBFL_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      echo $'/!\\ATTENTION: Nu W3C validator is too old./!\\'
      echo "Go get the latest version here:"
      echo "https://github.com/validator/validator/releases"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    declare -a LFBFL_files_for_Nu2
    mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
    readonly LFBFL_files_for_Nu2
    java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
      --verbose "${LFBFL_files_for_Nu2[@]}"
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.8 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# the GNU General Public License
```

```
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
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get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

is_top_dirstack_directory(){
  # This function should not be needed since when testing "${DIRSTACK[0]}"
  # is already a realpath. But I keep it, in case it is needed.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd requested: $1"
  declare -ig enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    enhanced_pushd_result=110
```

```
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  fi
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd validated1"
  is_top_dirstack_directory "$1" && {
    enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd validated2"

  pushd "$1" || {
    enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    echo "${LFBFL_where_was_i} $3$1 no such directory."
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd executed"
  # shellcheck disable=SC2248
  return ${enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_popd requested: $1"
  if [[ -z "$1" ]]; then
    return 0
  fi

  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    echo "${LFBFL_where_was_i} popd failed."
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_popd executed"
  return 0
```

```
}

can_continue_after_enhanced_pushd(){
  [[ enhanced_pushd_result -eq 0\
  || enhanced_pushd_result -eq 110\
  || enhanced_pushd_result -eq 111\
  ]]
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  # This test is faster than:
  #   for LFBFL_arg in "$@"; do
  #     if [[ "${LFBFL_arg}" == "--verbose" ]]; then
  # But it is not bug free.
  if [[ "$*" == *--verbose* ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    readonly LFBFL_i_popd_result
```

```
    echo "${LFBFL_where_was_i} $*"
    LFBFL_i_verbose=1
  fi
  # shellcheck disable=SC2034
  readonly LFBFL_i_verbose
}


# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --work-directory=* ]]; then
      LFBFL_work_directory=${LFBFL_arg#--work-directory=}
      LFBFL_work_directory=${LFBFL_work_directory:-.}
      # LFBFL_work_directory=${LFBFL_work_directory/#~/${HOME}} line below
      LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
      break
    fi
  done
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
}

popd_from_work_directory(){
  enhanced_popd "${LFBFL_work_directory}" 3
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}
```

```
# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
#       && trap 'enhanced_set_shell_option pipefail' RETURN
#   if you need to unset it in some function.
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  declare -r LFBFL_result_name="enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    echo "${LFBFL_where_was_i} $1 shell option activated."
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  declare -r LFBFL_result_name="enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set +o "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
```

```
    local LFBFL_where_was_i
    get_where_was_i 2
    echo "${LFBFL_where_was_i} $1 shell option unactivated."
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_set_bash_option(){
  # $1=optname some bash option for shopt
  declare -r LFBFL_result_name="enhanced_set_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
  # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
  if shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -s "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    echo "${LFBFL_where_was_i} $1 bash option activated."
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  declare -r LFBFL_result_name="enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    echo "${LFBFL_where_was_i} $1 bash option unactivated."
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
```

```
}
```

2.9 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
  # l'idée d'un principe de première différence ternaire.
  declare -i LFBFL_first=1
  local LFBFL_current
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ LFBFL_first -eq 1 ]]; then
      LFBFL_current="${LFBFL_arg}"
      LFBFL_first=0
```

```
      continue
    fi
    if [[ "${LFBFL_current}" != "${LFBFL_arg}" ]]; then
      return 0
    fi
  done
  return 1
}

some_distinct(){
  ! equal "$@"
}

all_distinct(){
  # Returns 1 if all the arguments strings are distinct.
  declare -ir LFBFL_ij_max=$#
  declare -i LFBFL_i
  declare -i LFBFL_j
  for ((LFBFL_i=1; LFBFL_i<=LFBFL_ij_max; ++LFBFL_i)); do
    for ((LFBFL_j=1; LFBFL_j<=LFBFL_ij_max; ++LFBFL_j)); do
      [[ LFBFL_i -eq LFBFL_j ]] && continue
      [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
    done
  done
  return 1
}

all_distinct2(){
  # Returns 1 if all the arguments strings are distinct.
  local LFBFL_element
  local LFBFL_previous_element
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  # Not setting pipefail since the result would still be incorrect.
  # shellcheck disable=SC2312
  printf "%s\n" "$@" \
    | sort\
    | {
      read -r LFBFL_previous_element\
      && while read -r LFBFL_element; do
        if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]];
        then
          return 0
        fi
        LFBFL_previous_element="${LFBFL_element}"
```

```
      done
      return 1
    }
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # This is the fastest of all 3; but note that version 2 is faster
  # than this one if a call to sort is added to this one.
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  # Not setting pipefail since the result would still be incorrect.
  # shellcheck disable=SC2162,SC2312
  printf "%s\n" "$@" \
    | uniq -d\
    | read
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
}

max(){
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
  # $1=$sort command with the flags you want to be applied.
  # $ max sort a b c
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1 -r"\
    | head -1
}
```

```
135
136 min(){
137   # Returns the minimum string among the arguments.
138   # Renvoie la chaîne la plus petite parmi celles proposées.
139   # $1=$sort command with the flags you want to be applied.
140   # $ min sort a b c
141   # a
142   if [[ $# -eq 0 ]]; then
143     return 1
144   fi
145   enhanced_set_shell_option pipefail\
146     && trap 'enhanced_unset_shell_option pipefail' RETURN
147   printf "%s\n" "${@:2}" \
148     | eval "$1"\
149     | head -1
150 }
151
152 is_substring(){
153   # $1=$string
154   # $2=$substring
155   declare -r LFBFL_var_1=$(
156     echo "$1"\
157     | sed -e 's/\\/\\&/g' -e "s/'/\\\\&/g"
158   )
159   declare -r LFBFL_var_2=$(
160     echo "$2"\
161     | sed -e 's/\\/\\&/g' -e "s/'/\\\\&/g"
162   )
163   local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
164   LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
165   LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
166   LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
167   readonly LFBFL_var_3
168   php -r "${LFBFL_var_3}"
169   return $?
170 }
171
172 is_subfile(){
173   # $1=$file_name
174   # $2=$sub_file_name
175   local LFBFL_var_1="\$a = file_get_contents('$1');"
176   LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
177   LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
178   LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
179   readonly LFBFL_var_1
180   php -r "${LFBFL_var_1}"
```

```
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

2.10 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
```

```
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  local LFBFL_subdir2="build_and_checks_variables"
  local LFBFL_data_file_name="${LFBFL_subdir2}/repository_data.txt"
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_repository_web_url=""
  grep_variable "${LFBFL_data_file_name}" repository_web_url\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_repository_git_url=""
  grep_variable "${LFBFL_data_file_name}" repository_git_url\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_abstract=""
  grep_variable "${LFBFL_data_file_name}" abstract\
    --result-variable-prefix="LFBFL_"

  local LFBFL_acknowledgments=""
  grep_variable "${LFBFL_data_file_name}" acknowledgments\
    --result-variable-prefix="LFBFL_"

  local LFBFL_author_email=""
  grep_variable "${LFBFL_data_file_name}" author_email\
    --result-variable-prefix="LFBFL_"\
```

```
85      --replace-line-returns-by=""
86
87    local LFBFL_author_full_name=""
88    grep_variable "${LFBFL_data_file_name}" author_full_name\
89      --result-variable-prefix="LFBFL_"\
90      --replace-line-returns-by=" "
91
92    local LFBFL_author_website=""
93    grep_variable "${LFBFL_data_file_name}" author_website\
94      --result-variable-prefix="LFBFL_"\
95      --replace-line-returns-by=""
96
97    declare -i LFBFL_max_line_length
98    grep_variable "${LFBFL_data_file_name}" max_line_length\
99      --result-variable-prefix="LFBFL_"
100
101   declare -r LFBFL_current_date=$(date -I"date")
102
103   declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)
104
105   declare -ir LFBFL_number_of_commits=$(
106     git shortlog\
107     | space_starting_lines\
108     | wc -l
109   )
110
111   local LFBFL_redirect="/dev/stdout"
112   if [[ LFBFL_i_verbose -eq 0 ]]; then
113     LFBFL_redirect="/dev/null"
114   fi
115   {
116     code_lines_count_all "$@"
117     code_lines_count_not_empty "$@"
118     code_lines_count_empty "$@"
119   } &> "${LFBFL_redirect}"
120
121   local LFBFL_number_of_lines="${code_lines_count_all_result}"
122   LFBFL_number_of_lines+=" total lines,"
123   LFBFL_number_of_lines+=" ${code_lines_count_not_empty_result}"
124   LFBFL_number_of_lines+=" not empty lines,"
125   LFBFL_number_of_lines+=" ${code_lines_count_empty_result}"
126   LFBFL_number_of_lines+=" empty lines."
127   readonly LFBFL_number_of_lines
128
129   local LFBFL_temp_path
130   # LFBFL_temp_path=$(realpath "./${LFBFL_subdir2}/temp") for pushd
```

```
LFBFL_temp_path="./${LFBFL_subdir2}/temp"
readonly LFBFL_temp_path

# see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
# ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
# ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
tree -a --gitignore\
  -I "node_modules/"\
  -I "__pycache__/"\
  -I ".mypy_cache/"\
  -I ".ruff_cache/"\
  -I ".git/"\
  | replace_non_ascii_spaces\
  > "${LFBFL_temp_path}/current_tree_light.txt"

tree -a -DFh --gitignore\
  -I "node_modules/"\
  -I "__pycache__/"\
  -I ".mypy_cache/"\
  -I ".ruff_cache/"\
  -I ".git/"\
  | replace_non_ascii_spaces\
  > "${LFBFL_temp_path}/current_tree.txt.temp"

local LFBFL_s_file_paths
LFBFL_s_file_paths=$(
  sed -Ez 's/(.)\\\n/\1/Mg' "./${LFBFL_subdir2}/files_names_listing.txt"\
  | grep -v '^//'
)
readonly LFBFL_s_file_paths
declare -a LFBFL_arr_file_paths
mapfile -t LFBFL_arr_file_paths <<< "${LFBFL_s_file_paths}"
readonly LFBFL_arr_file_paths

local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
LFBFL_temp_files_listing+="files_listing.tex.tpl.temp"
readonly LFBFL_temp_files_listing
: > "${LFBFL_temp_files_listing}"
local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
LFBFL_temp_files_listing2+="files_listing.html.tpl.temp"
readonly LFBFL_temp_files_listing2
: > "${LFBFL_temp_files_listing2}"
# HTML <li> elements, hence "lis".
local LFBFL_temp_files_lis
LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.tpl"
readonly LFBFL_temp_files_lis
```

```
  : > "${LFBFL_temp_files_lis}"
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_max_line_length} /
  declare -r LFBFL_score_command="${get_split_score_result}"
  local LFBFL_score_command_properties="${get_split_score_result2}"
  readonly LFBFL_score_command_properties
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_max_line_length} ':'
  declare -r LFBFL_score_command2="${get_split_score_result}"
  local LFBFL_score_command_properties2="${get_split_score_result2}"
  readonly LFBFL_score_command_properties2
  declare -r LFBFL_suffix='%'
  declare -i LFBFL_i=0
  local LFBFL_cleaned_path1
  local LFBFL_cleaned_path2
  local LFBFL_new_lines
  local LFBFL_new_lines2
  local LFBFL_new_lines3
  local LFBFL_file_path
  # Remove line returns here to keep lines short.
  for LFBFL_file_path in "${LFBFL_arr_file_paths[@]}"; do
    printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
    # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
    # LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
    LFBFL_cleaned_path2=$(
      printf "%s" "${LFBFL_file_path}"\
      | sed -e 's/\//:/g' -e 's/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
```

```
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        "${LFBFL_score_command_properties2}"
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_path}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    {
      printf "\\subsection{\n"
      printf "%s\n" "${LFBFL_new_lines}"
      printf "}\n"
      printf "\\label{\n"
      printf "%s\n" "${LFBFL_new_lines2}"
      printf "}\n"
      printf "\n"
      printf "\\VerbatimInput[numbers=left,xleftmargin=-5mm]{\n"
      printf "%s\n" "${LFBFL_new_lines3}"
      printf "}\n"
      printf "\n"
      printf "\n"
    } >> "${LFBFL_temp_files_listing}"

    # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_path}"
    if [[ ${#LFBFL_file_path} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
```

```
      printf "<h3 id=\"subsection2.%s\">2.%s\n" "${LFBFL_i}" "${LFBFL_i}"
      printf "%s\n" "${LFBFL_new_lines}"
      printf "</h3>\n"
      printf "<pre class=\"numbered_lines\">\n"
      sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g'\
        < "${LFBFL_file_path}"
      printf "</pre>\n"
      printf "\n"
      printf "\n"
    } >> "${LFBFL_temp_files_listing2}"
    {
      printf "      <li><a href=\"#subsection2.%s\">\n" "${LFBFL_i}"
      printf "%s\n" "${LFBFL_new_lines}"
      printf "      </a></li>\n"
    } >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "./${LFBFL_subdir2}/files_listing.tex.tpl"\
    "${LFBFL_temp_files_listing}"
  sed -i -e "s|\\n</pre>|</pre>|g" "${LFBFL_temp_files_listing2}"
  overwrite_if_not_equal\
    "${LFBFL_temp_path}/files_listing.html.tpl"\
    "${LFBFL_temp_files_listing2}"

  # We verify if some lines are beyond max_line_length characters
  # in current_tree_light.txt and current_tree.txt.
  declare -ar LFBFL_trees=(
    "current_tree_light.txt"
    "current_tree.txt.temp"
  )
  local LFBFL_tree_path
  local LFBFL_line
  local LFBFL_prefix
  declare -i LFBFL_prefix_last_position
  local LFBFL_char
  local LFBFL_line_start
  declare -ir LFBFL_overlength=$((LFBFL_max_line_length+1))
  local LFBFL_file_name
  local LFBFL_s_lines
  declare -a LFBFL_arr_lines
  declare -i LFBFL_keep_length
  for LFBFL_tree in "${LFBFL_trees[@]}"; do
    LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
    # shellcheck disable=SC2248
    LFBFL_s_lines=$(grep '.\{'${LFBFL_overlength}'\}' "${LFBFL_tree_path}")
    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
    for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
```

```
      # printf "LFBFL_line: %s\n" "${LFBFL_line}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E -e 's/(.*)──[^─]+$/\1/g'
      )
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_prefix_last_position=$((${#LFBFL_prefix}-1))
      LFBFL_char="${LFBFL_prefix:${LFBFL_prefix_last_position}:1}"
      # printf "LFBFL_char: %s\n" "${LFBFL_char}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_prefix}"\
        | sed -E -e 's/[^ ]+$//g'
      )
      if [[ "${LFBFL_char}" == "└" ]]; then
        LFBFL_prefix+="  "
      else
        LFBFL_prefix+="│ "
      fi
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_file_name=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E 's|.* (([a-zA-Z0-9\._/-]+).)$|\1|g'
      )
      # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
      LFBFL_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
      LFBFL_line_start=${LFBFL_line:0:${LFBFL_keep_length}}
      LFBFL_line_start=$(
        printf "%s" "${LFBFL_line_start}"\
        | sed -e 's/ *$//g'
      )
      # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
        > "${LFBFL_tree_path}.some_line.temp"
      insert_file_at_token "${LFBFL_tree_path}"\
        "${LFBFL_line}"\
```

```
        "${LFBFL_tree_path}.some_line.temp"
    done
    rm -f "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_temp_path}/current_tree.txt.temp" 1 1

  declare -r LFBFL_tex_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
  declare -r LFBFL_html_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.html"

  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]];
  then
    # If there is a template, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]];
  then
    # Otherwise if there is a tex file, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
    printf "Neither .tex.tpl, nor .tex in ./%s/\n" "${LFBFL_subdir2}"
  fi

  # Same logic with repository HTML file.
  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]];
  then
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
  elif [[ -f "./${LFBFL_repository_name}.html" ]]; then
    cp "./${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    printf "Neither .html.tpl in ./%s/, nor .html in ./\n"\
      "${LFBFL_subdir2}"
  fi

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_abstract}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/abstract_temp"

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_acknowledgments}"\
```

```
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/acknowledgments_temp"

  # But the filling still occurs, in case the dev want to refill
  # part of the tex/html file that he modified with @token@
  # using some computed results.
  # Tex filling:
  if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
           -e "s|@number_of_commits@|${LFBFL_number_of_commits}|g"\
           -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_tex_path_start}.1"

    insert_file_at_token "${LFBFL_tex_path_start}.1" @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_tex_path_start}.2"

    insert_file_at_token "${LFBFL_tex_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_tex_path_start}.3"

    insert_file_at_token "${LFBFL_tex_path_start}.3"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_tex_path_start}.4"

    insert_file_at_token "${LFBFL_tex_path_start}.4"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_tex_path_start}.5"

    insert_file_at_token "${LFBFL_tex_path_start}.5"\
      @files_listing_VerbatimInput@\
      "./${LFBFL_subdir2}/files_listing.tex.tpl"\
      "${LFBFL_tex_path_start}.6"

    overwrite_if_not_equal\
      "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
```

```
453         "${LFBFL_tex_path_start}.6" 1
454     fi
455
456     # HTML filling:
457     declare -i LFBFL_HTML_updated=0
458     if [[ -f "${LFBFL_html_path_start}.1" ]]; then
459       sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
460              -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
461              -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
462              -e "s|@author_email@|${LFBFL_author_email}|g"\
463              -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
464              -e "s|@author_website@|${LFBFL_author_website}|g"\
465              -e "s|@current_date@|${LFBFL_current_date}|g"\
466              -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
467              -e "s|@number_of_commits@|${LFBFL_number_of_commits}|g"\
468              -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
469         "${LFBFL_html_path_start}.1"
470
471       insert_file_at_token "${LFBFL_html_path_start}.1" @abstract@\
472         "${LFBFL_temp_path}/abstract_temp"\
473         "${LFBFL_html_path_start}.2"
474
475       insert_file_at_token "${LFBFL_html_path_start}.2"\
476         @acknowledgments@\
477         "${LFBFL_temp_path}/acknowledgments_temp"\
478         "${LFBFL_html_path_start}.3"
479
480       insert_file_at_token "${LFBFL_html_path_start}.3"\
481         @files_lis@ "${LFBFL_temp_files_lis}"\
482         "${LFBFL_html_path_start}.4"
483
484       insert_file_at_token "${LFBFL_html_path_start}.4"\
485         @current_tree_light@\
486         "${LFBFL_temp_path}/current_tree_light.txt"\
487         "${LFBFL_html_path_start}.5"
488
489       insert_file_at_token "${LFBFL_html_path_start}.5"\
490         @current_tree@\
491         "${LFBFL_temp_path}/current_tree.txt"\
492         "${LFBFL_html_path_start}.6"
493
494       insert_file_at_token "${LFBFL_html_path_start}.6"\
495         @files_listing_HTMLPreInput@\
496         "${LFBFL_temp_path}/files_listing.html.tpl"\
497         "${LFBFL_html_path_start}.7"
498
```

```
    overwrite_if_not_equal "./${LFBFL_repository_name}.html"\
      "${LFBFL_html_path_start}.7" 1
    LFBFL_HTML_updated=$?
  fi

  # The HTML contains everything including listings with source code.
  # If it doesn't get updated,
  # then it is useless to recompute the PDF.
  if [[ LFBFL_HTML_updated -eq 0 ]]; then
    return
  fi

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    for ((i=0; i<3; i++)); do
      pdflatex "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"
    done
  else
    for ((i=0; i<3; i++)); do
      pdflatex "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
        > /dev/null
    done
  fi

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

2.11 build_and_checks_dependencies/generate_from_template.libr.sh

```
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#
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```

```
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#
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_name
  # $2=target_file_name
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    echo "$3"
    cat "$1"
    echo "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
  # $1=base_file_name
  # $2=target_file_name
  # $3=line_comment_prefix
  # $4=optional_post_processing
  # $5=file_prefix
```

```
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
  readonly LFBFL_temp
  if [[ -n "$5" ]]; then
    echo "$5"\
      > "${LFBFL_temp}"
  fi
  sed -e "s/^/$3/g" "$1"\
    >> "${LFBFL_temp}"
  if [[ -n "$4" ]]; then
    eval "$4"
  fi
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

split_file_in_two(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_part1
  # $4=$file_name_part2
  # Other arguments are options for grep.
  declare -a LFBFL_grep_options=()
  declare -i LFBFL_i=0
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    ((++LFBFL_i))
    if [[ LFBFL_i -lt 5 ]]; then
      continue
    fi
    LFBFL_grep_options+=("${LFBFL_arg}")
  done
  declare -ir LFBFL_line_number=$(
    grep -n "${LFBFL_grep_options[@]}" -- "$2" "$1"\
    | head --lines=1\
    | cut -f 1 -d ':'
  )
  declare -ir LFBFL_line_count=$(ll_wc -l -n "$1")
  declare -ir LFBFL_lines_after=$((
    LFBFL_line_count - LFBFL_line_number
  ))
  head --lines="$((LFBFL_line_number - 1))" "$1"\
    > "$3"
  tail --lines="${LFBFL_lines_after}" "$1"\
    > "$4"
}
```

```
insert_file_at_token(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_to_insert
  # $4=$file_name_result
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
    "${LFBFL_end_file_name}" --fixed-strings
  cat "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}"
  fi
}
```

2.12 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
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```
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# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
```

```
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

2.13 build_and_checks_dependencies/grammar_and_spell_check.libr.sh

```
#!/usr/bin/env bash
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```

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# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spell_check(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_grammar_or_spell_checker_command=""
  grep_variable "$1" grammar_or_spell_checker_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_or_spell_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    return
  fi
  local LFBFL_file_path
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 0 ]]\
```

```
      || echo "Iterating on pattern: ${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | while read -r LFBFL_file_path;
    do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval "${LFBFL_eval_string}"
    done
  done
}
```

2.14 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
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# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
```

```
34 # shellcheck source=generate_from_template.libr.sh
35 source "./${LFBFL_subdir}/generate_from_template.libr.sh"
36 # shellcheck source=lines_filters.libr.sh
37 source "./${LFBFL_subdir}/lines_filters.libr.sh"
38 # shellcheck source=overwrite_if_not_equal.libr.sh
39 source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
40
41 build_licenses_templates(){
42   # Options:
43   #   --verbose
44   #   --work-directory=""
45   declare -i LFBFL_i_verbose=0
46   get_verbose_option "$@"
47   local LFBFL_work_directory=""
48   get_work_directory_option "$@"
49
50   enhanced_set_shell_option pipefail\
51     && trap 'enhanced_unset_shell_option pipefail' RETURN
52
53   # Source
54   local LFBFL_license_source_subdir
55   LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
56   readonly LFBFL_license_source_subdir
57   local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
58   LFBFL_source_license_prefix+="license_file_header_"
59   readonly LFBFL_source_license_prefix
60   # Target
61   local LFBFL_license_target_subdir
62   LFBFL_license_target_subdir="${LFBFL_work_directory}/"
63   LFBFL_license_target_subdir="build_and_checks_variables/temp/"
64   readonly LFBFL_license_target_subdir
65   local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
66   LFBFL_target_license_prefix+="license_file_header_"
67   readonly LFBFL_target_license_prefix
68
69   declare -ar LFBFL_licenses=(
70     "AGPLv3+"
71     "GPLv3+"
72     "LGPLv3+"
73   )
74
75   # ------------------------------------------------------------------
76   # Various language definitions for text file formats
77   declare -ar LFBFL_block_comment_languages=(
78     "c"
79     "py"
```

```
)
declare -ar LFBFL_block_comment_enters=(
  '/*'
  '"""'
)
declare -ar LFBFL_block_comment_exits=(
  '*/'
  '"""'
)
equal "${#LFBFL_block_comment_languages[@]}"\
      "${#LFBFL_block_comment_enters[@]}"\
      "${#LFBFL_block_comment_exits[@]}"
declare -i LFBFL_equal_result=$?
if [[ LFBFL_equal_result -eq 0 ]]; then
  echo $'/!\\Problem with the definitions of languages arrays 1/!\\'
fi

declare -ar LFBFL_line_comment_languages=(
  "sh"
  "tex"
)
declare -ar LFBFL_line_comment_prefixes=(
  '# '
  '% '
)
equal "${#LFBFL_line_comment_languages[@]}"\
      "${#LFBFL_line_comment_prefixes[@]}"
LFBFL_equal_result=$?
if [[ LFBFL_equal_result -eq 0 ]]; then
  echo $'/!\\Problem with the definitions of languages arrays 2/!\\'
fi
# ------------------------------------------------------------------

# ------------------------------------------------------------------
# We generate the license header templates adapted to various text
# files formats.
# Kamoulox do endfor, Jacques Beauregard XD
# (C'est toujours mieux que l'inverse :) !)
# for do endfor -> for do done (Fort Doux Donne).
local LFBFL_some_license
local LFBFL_license_prefix
local LFBFL_license_prefix2
local LFBFL_extension
local LFBFL_license_file_name
local LFBFL_enter_string
local LFBFL_exit_string
```

```
126     local LFBFL_temp2
127     local LFBFL_prefix_string
128     local LFBFL_intermediate_file_name
129     local LFBFL_file_prefix
130     declare -i LFBFL_generate_from_template_result
131     for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
132       LFBFL_license_prefix="${LFBFL_source_license_prefix}"
133       LFBFL_license_prefix+="${LFBFL_some_license}"
134       LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
135       LFBFL_license_prefix2+="${LFBFL_some_license}"
136       for ((i=0; i<${#LFBFL_block_comment_languages[@]}; i++)); do
137         LFBFL_extension=${LFBFL_block_comment_languages[i]}
138         LFBFL_license_file_name="${LFBFL_license_prefix2}"
139         LFBFL_license_file_name+=".${LFBFL_extension}"
140         LFBFL_enter_string=${LFBFL_block_comment_enters[i]}
141         LFBFL_exit_string=${LFBFL_block_comment_exits[i]}
142         generate_from_template_with_block_comments\
143           "${LFBFL_license_prefix}.tpl"\
144           "${LFBFL_license_file_name}.tpl"\
145           "${LFBFL_enter_string}" "${LFBFL_exit_string}"
146         LFBFL_generate_from_template_result=$?
147         if [[ LFBFL_i_verbose -eq 1 ]]; then
148           if [[ LFBFL_generate_from_template_result -eq 1 ]]; then
149             echo\
150               "License template ${LFBFL_license_file_name}.tpl updated"
151           elif [[ LFBFL_generate_from_template_result -eq 2 ]]; then
152             echo\
153               "License template ${LFBFL_license_file_name}.tpl created"
154           fi
155         fi
156       done
157       LFBFL_temp2=".generate_from_template_with_line_comments.temp"
158       for ((i=0; i<${#LFBFL_line_comment_languages[@]}; i++)); do
159         LFBFL_extension=${LFBFL_line_comment_languages[i]}
160         LFBFL_license_file_name="${LFBFL_license_prefix2}"
161         LFBFL_license_file_name+=".${LFBFL_extension}"
162         LFBFL_prefix_string=${LFBFL_line_comment_prefixes[i]}
163         LFBFL_intermediate_file_name="${LFBFL_license_file_name}.tpl"
164         LFBFL_intermediate_file_name+="${LFBFL_temp2}"
165         LFBFL_file_prefix=""
166         if [[ "${LFBFL_extension}" == "sh" ]]; then
167           LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
168         fi
169         # The sed command is here to remove trailing spaces of previously
170         # empty lines after comment prefix is added to the line.
171         generate_from_template_with_line_comments\
```

```
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_name}.tpl"\
        "${LFBFL_prefix_string}"\
        "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_name}'"\
        "${LFBFL_file_prefix}"
      LFBFL_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_generate_from_template_result -eq 1 ]]; then
          echo\
            "License template ${LFBFL_license_file_name}.tpl updated"
        elif [[ LFBFL_generate_from_template_result -eq 2 ]]; then
          echo\
            "License template ${LFBFL_license_file_name}.tpl created"
        fi
      fi
    done
  done
  # ------------------------------------------------------------------

  # Now that base license templates generated languages license
  # templates in working directory, we can go there.
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  local LFBFL_data_file_name="build_and_checks_variables/"
  LFBFL_data_file_name+="repository_data.txt"
  readonly LFBFL_data_file_name
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  local LFBFL_license=""
  grep_variable "${LFBFL_data_file_name}" license\
    --result-variable-prefix="LFBFL_"
  local LFBFL_license2=""
  grep_variable "${LFBFL_data_file_name}" license2\
    --result-variable-prefix="LFBFL_"
  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_name}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "

  LFBFL_license_prefix="build_and_checks_variables/temp/"
  LFBFL_license_prefix+="license_file_header_"
  readonly LFBFL_license_prefix
```

```
LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
readonly LFBFL_license_prefix2
declare -r\
  LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"

# First year according to current state of git repository.
declare -ir LFBFL_first_year=$(
  git log\
  | grep 'Date:'\
  | cut -f 8 -d ' '\
  | tail -1
)
# Last year according to current state of git repository.
declare -ir LFBFL_last_year=$(
  git log\
  | grep 'Date:'\
  | cut -f 8 -d ' '\
  | head -1
)
local LFBFL_copyright_string
LFBFL_copyright_string="${LFBFL_first_year}-${LFBFL_last_year}"
LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
readonly LFBFL_copyright_string

prepare_filled_license_file_block(){
  # $1=license_file_name
  sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
      -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
      "$1.tpl"\
    > "$1.temp"
  overwrite_if_not_equal "$1" "$1.temp"
  declare -r LFBFL_result=$?
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    if [[ LFBFL_result -eq 1 ]]; then
      echo "License template $1 updated"
    elif [[ LFBFL_result -eq 2 ]]; then
      echo "License template $1 created"
    fi
  fi
  head --lines=-1 "$1"\
    | tail --lines=+2\
    > "$1.temp"
  # Strange world: tail --lines=-1 was working
  # to exclude first line.
  # I tested it months ago, and saw with my eyes the start and end
  # block comments removed with:
```

```
    # head --lines=-1 "$1" | tail --lines=-1.
    # Now, man page says to use --lines=+2 instead.
    # Welcome in the world where AI change your environment and your
    # documentation on the fly...
  }

  declare -Ar LFBFL_all_block_comment_languages=(
    ["c"]="c"
    ["php"]="c"
    ["py"]="py"
  )

  local LFBFL_key
  local LFBFL_dest
  local LFBFL_license_file_name2
  declare -i LFBFL_not_subfile
  declare -i LFBFL_not_subfile2
  for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_block "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_block "${LFBFL_license_file_name2}"
    fi
    find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
      | relevant_find\
      | while read -r LFBFL_file_name;
    do
      is_subfile "${LFBFL_file_name}"\
        "${LFBFL_license_file_name}.temp"
      LFBFL_not_subfile=$?
      LFBFL_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_name}"\
          "${LFBFL_license_file_name2}.temp"
        LFBFL_not_subfile2=$?
      fi
      if [[ LFBFL_not_subfile -ge 1 && LFBFL_not_subfile2 -ge 1 ]];
      then
        echo "File ${LFBFL_file_name} has no/wrong license header?"
      fi
    done
    rm "${LFBFL_license_file_name}.temp"
    if [[ -n "${LFBFL_license2}" ]]; then
```

```
      rm "${LFBFL_license_file_name2}.temp"
    fi
  done

  prepare_filled_license_file_line(){
    # $1=LFBFL_license_file_name
    sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
        -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
        "$1.tpl"\
      > "$1.temp"
    overwrite_if_not_equal "$1" "$1.temp"
    declare -r LFBFL_result=$?
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ LFBFL_result -eq 1 ]]; then
        echo "License template $1 updated"
      elif [[ LFBFL_result -eq 2 ]]; then
        echo "License template $1 created"
      fi
    fi
  }

  declare -Ar LFBFL_all_line_comment_languages=(
    ["sh"]="sh"
    ["tex"]="tex"
  )

  for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_line "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_line "${LFBFL_license_file_name2}"
    fi
    find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
      | relevant_find\
      | while read -r LFBFL_file_name;
    do
      is_subfile "${LFBFL_file_name}" "${LFBFL_license_file_name}"
      LFBFL_not_subfile=$?
      LFBFL_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_name}" "${LFBFL_license_file_name2}"
        LFBFL_not_subfile2=$?
      fi
```

```
      if [[ LFBFL_not_subfile -ge 1 && LFBFL_not_subfile2 -ge 1 ]];
      then
        echo "File ${LFBFL_file_name} has no/wrong license header?"
      fi
    done
  done
}

build_licenses_templates "$@"
```

2.15 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

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```

2.16 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

2.17 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

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```

2.18 build_and_checks_dependencies/lines_counts.libr.sh

```
#!/usr/bin/env bash
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#
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```

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#
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# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
```

```
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && echo "Iterating on pattern: ${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H -v 'a(?!a)a'
  done
}

all_self_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  # below and another lines, see SC2145 false positive
  all_code_lines "$@" \
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  all_self_code_lines "$@" \
    | empty_lines_after_file_name
}

```

```
# shellcheck disable=SC2120
all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  all_self_code_lines "$@" \
    | not_empty_lines_after_file_name
}

code_lines_count_all(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi code_lines_count_all_result
  code_lines_count_all_result=$(
    all_self_code_lines\
    | wc -l
  )
  # The echo is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  echo ${code_lines_count_all_result}
}

code_lines_count_empty(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

```

```
  declare -gi code_lines_count_empty_result
  # shellcheck disable=SC2119
  code_lines_count_empty_result=$(
    all_self_empty_code_lines\
    | wc -l
  )
  # The echo is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  echo ${code_lines_count_empty_result}
}

code_lines_count_not_empty(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc -l
  )
  # The echo is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  echo ${code_lines_count_not_empty_result}
}
```

2.19 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
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```

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# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_number_only=0
  declare -i ll_wc_var_i=0
```

```
  for ll_wc_var_arg in "$@"; do
    # echo "$ll_wc_var_arg"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_number_only=1
    elif [[ ll_wc_var_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # echo "$ll_wc_var_number_only"
  if [[ ll_wc_var_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # echo "normal"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# echo "https://domain/list-path/2024-05/msg00013.html" | \
#   sed -e 's|domain|lists.gnu.org|' \
#       -e 's|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
```

```
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
  declare -ir LFBFL_lines_before=$(ll_wc -l -n "${!#}")
  declare -ir LFBFL_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # echo "$LFBFL_lines_before"
  # echo "$LFBFL_lines_after"
  if [[ "${LFBFL_lines_before}" ==  "${LFBFL_lines_after}" ]]; then
    rm "${LFBFL_temp}"
    return
  fi
  mv "${LFBFL_temp}" "${!#}"
}

grep_fixed_string_with_anchor(){
  # $1=filepath
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below, you don't need to
  #   supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
  declare -i LFBFL_enforce_line_starts=0
  declare -i LFBFL_enforce_line_ends=0
  declare -i LFBFL_quiet=0
```

```
149     local LFBFL_arg
150     for LFBFL_arg in "$@"; do
151       ((++LFBFL_i))
152       if [[ LFBFL_i -lt 3 ]]; then
153         continue
154       fi
155       if [[ "${LFBFL_arg}" == "--enforce-line-starts-with-fixed-string" ]];
156       then
157         LFBFL_enforce_line_starts=1
158         continue
159       fi
160       if [[ "${LFBFL_arg}" == "--enforce-line-ends-with-fixed-string" ]];
161       then
162         LFBFL_enforce_line_ends=1
163         continue
164       fi
165       if [[ "${LFBFL_arg}" == "-q" || "${LFBFL_arg}" == "--quiet" ]];
166       then
167         LFBFL_quiet=1
168         LFBFL_grep_options+=("${LFBFL_arg}")
169         continue
170       fi
171       LFBFL_grep_options2+=("${LFBFL_arg}")
172     done
173
174     if [[
175       LFBFL_enforce_line_starts -eq 0
176       && LFBFL_enforce_line_ends -eq 0
177       ]];
178     then
179       grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
180       return
181     fi
182     declare -r LFBFL_s_lines=$(
183       grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
184         "$2" "$1"
185     )
186     declare -a LFBFL_arr_lines
187     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
188     readonly LFBFL_arr_lines
189
190     declare -i LFBFL_fixed_string_length=${#2}
191     declare -i LFBFL_i_result=1
192     local LFBFL_some_line
193     local LFBFL_line_part
194     for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
```

```
    if [[ "${LFBFL_some_line}" == "$2" ]]; then
      [[ LFBFL_quiet -eq 1 ]] || printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_result=0
      continue
    fi
    if [[
      LFBFL_enforce_line_starts -eq 1
      && LFBFL_enforce_line_ends -eq 1
      ]];
    then
      continue
    fi
    if [[ LFBFL_enforce_line_starts -eq 1 ]]; then
      LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_fixed_string_length}}
    fi
    if [[ LFBFL_enforce_line_ends -eq 1 ]]; then
      LFBFL_line_part=${LFBFL_some_line: -${LFBFL_fixed_string_length}}
    fi
    if [[ "${LFBFL_line_part}" == "$2" ]]; then
      [[ LFBFL_quiet -eq 1 ]] || printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_result=0
    fi
  done
  LFBFL_i_result=$((LFBFL_i_result | $?))
  # shellcheck disable=SC2248
  return ${LFBFL_i_result}
}

grep_variable(){
  # $1=$file
  # $2=$variable_name
  # Options :
  #   --replace-line-returns-by=""
  #   --result-variable-prefix="LFBFL_" for example
  declare -r LFBFL_regexp="(?<=^$2=).*$"
  local LFBFL_arg
  # echo $LFBFL_regexp
  local LFBFL_variable_value
  if [[ "$*" == *--replace-line-returns-by=* ]]; then
    local LFBFL_s_replace=""
    for LFBFL_arg in "$@"; do
      if [[ "${LFBFL_arg}" == --replace-line-returns-by=* ]]; then
        LFBFL_s_replace=${LFBFL_arg#--replace-line-returns-by=}
        break
      fi
    done
```

```
    LFBFL_variable_value=$(
      grep -oP "${LFBFL_regexp}" "$1"\
      | sed -Ez -e 's/\n/'"${LFBFL_s_replace}"'/Mg'
    )
    # We remove the trailing replace string,
    # since the intent is only to replace the intermediate \n.
    if [[ -n "${LFBFL_s_replace}" ]]; then
    LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
    fi
  else
    LFBFL_variable_value=$(grep -oP "${LFBFL_regexp}" "$1")
  fi
  local LFBFL_prefix=""
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --result-variable-prefix=* ]]; then
      LFBFL_prefix=${LFBFL_arg#--result-variable-prefix=}
      break
    fi
  done
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # echo "${LFBFL_variable_name} : ${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # echo "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep -v '^$'
}

space_starting_lines(){
  grep '^ '
}

not_space_starting_lines(){
  grep -v '^ '
}

# Space starting or empty lines = spampty
```

```
spampty_lines(){
  grep -E '^( |$)'
}

not_spampty_lines(){
  grep '^[^ ]'
  # grep -v -E '^( |$)'
}

all_space_lines(){
  grep '^[ ]\+$'
}

not_all_space_lines(){
  grep -v '^[ ]\+$'
}

# All space or empty lines = aspampty
aspampty_lines(){
  grep '^[ ]*$'
}

not_aspampty_lines(){
  grep '[^ ]'
  # grep -v '^[ ]*$'
}

empty_lines_after_file_name(){
  grep '^[^:]\+:$'
}

not_empty_lines_after_file_name(){
  grep -v '^[^:]\+:$'
}

space_starting_lines_after_file_name(){
  grep '^[^:]\+: '
}

not_space_starting_lines_after_file_name(){
  grep -v '^[^:]\+: '
}

spampty_lines_after_file_name(){
  grep -E '^[^:]+:( |$)'
}
```

```
not_spampty_lines_after_file_name(){
  # The two variants results differ on lines without filename before.
  # Since the intent of the function is only on lines "blabla.sh:whatever",
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep -v -E '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}

not_all_space_lines_after_file_name(){
  grep -v '^[^:]\+:[ ]\+$'
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep -v '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# :). Nice command :).

not_JS_dependencies_find(){
  grep -vE -e "(^|/)node_modules/"\
           -e "(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep -vE -e "^([^:]+/)?node_modules/"\
           -e "^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}
```

```
379
380 not_dependencies_grep(){
381   not_JS_dependencies_grep
382 }
383
384 not_python_cache_find(){
385   grep -vE -e "(^|/)__pycache__/"\
386            -e "(^|/)\.mypy_cache/"\
387            -e "(^|/)\.ruff_cache/"
388 }
389
390 not_python_cache_grep(){
391   grep -vE -e "^([^:]+/)?__pycache__/"\
392            -e "^([^:]+/)?\.mypy_cache/"\
393            -e "^([^:]+/)?\.ruff_cache/"
394 }
395
396 not_cache_find(){
397   not_python_cache_find
398 }
399
400 not_cache_grep(){
401   not_python_cache_grep
402 }
403
404 not_git_find(){
405   grep -vE "(^|/)\.git/"
406 }
407
408 not_git_grep(){
409   grep -vE "^([^:]+/)?\.git/"
410 }
411
412 not_archive_find(){
413   grep -vE "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
414 }
415
416 not_archive_grep(){
417   grep -vE "^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
418 }
419
420 not_license_find(){
421   grep -vE -e "(^|/)COPYING$"\
422            -e "(^|/)COPYING\.LESSER$"
423 }
424
```

```
not_license_grep(){
  grep -vE -e "^([^:]+/)?COPYING:"\
           -e "^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.html$"
}

not_main_html_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.html:"
}

not_temp_file_find(){
  grep -vE "(^|/)build_and_checks_variables/temp/.*$"
}

not_temp_file_grep(){
```

```
  grep -vE "^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
}

relevant_find(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_find\
    | not_cache_find\
    | not_git_find\
    | not_archive_find\
    | not_temp_file_find
}

relevant_grep(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep
}
```

2.20 build_and_checks_dependencies/lines_maps.libr.sh

```
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```

```
#
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# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed -e 's/\xc2\xa0/ /g'\
      -e 's/\xe2\x80\x80/ /g'\
      -e 's/\xe2\x80\x81/ /g'\
      -e 's/\xe2\x80\x82/ /g'\
      -e 's/\xe2\x80\x83/ /g'\
      -e 's/\xe2\x80\x84/ /g'\
      -e 's/\xe2\x80\x85/ /g'\
      -e 's/\xe2\x80\x86/ /g'\
      -e 's/\xe2\x80\x87/ /g'\
      -e 's/\xe2\x80\x88/ /g'\
      -e 's/\xe2\x80\x89/ /g'\
      -e 's/\xe2\x80\x8a/ /g'\
      -e 's/\xe2\x80\x8b/ /g'\
      -e 's/\xe2\x80\x8c/ /g'\
      -e 's/\xe2\x80\x8d/ /g'\
      -e 's/\xe2\x80\x8e/ /g'\
      -e 's/\xe2\x80\x8f/ /g'\
      -e 's/\xe2\x80\xa8/ /g'\
      -e 's/\xe2\x80\xa9/ /g'\
      -e 's/\xe2\x80\xaa/ /g'\
      -e 's/\xe2\x80\xab/ /g'\
      -e 's/\xe2\x80\xac/ /g'\
      -e 's/\xe2\x80\xad/ /g'\
      -e 's/\xe2\x80\xae/ /g'\
      -e 's/\xe2\x80\xaf/ /g'\
      -e 's/\xe2\x81\x9f/ /g'\
      -e 's/\xe2\x81\xa0/ /g'\
      -e 's/\xe2\x81\xa1/ /g'\
      -e 's/\xe2\x81\xa2/ /g'\
      -e 's/\xe2\x81\xa3/ /g'\
```

```
      -e 's/\xe2\x81\xa4/ /g'\
      -e 's/\xe2\x81\xa5/ /g'\
      -e 's/\xe2\x81\xa6/ /g'\
      -e 's/\xe2\x81\xa7/ /g'\
      -e 's/\xe2\x81\xa8/ /g'\
      -e 's/\xe2\x81\xa9/ /g'\
      -e 's/\xe2\x81\xaa/ /g'\
      -e 's/\xe2\x81\xab/ /g'\
      -e 's/\xe2\x81\xac/ /g'\
      -e 's/\xe2\x81\xad/ /g'\
      -e 's/\xe2\x81\xae/ /g'\
      -e 's/\xe2\x81\xaf/ /g'
}
```

2.21 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
  # $1=$target_file_name
  # $2=$temp_file_name
  # $3=$keep_temp
```

```
  # $4=$tree_mode : when the two files are the output of tree command,
  # we ignore if all the differences are only on directories.
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  if [[ ! -f "$1" ]]; then
    if [[ -n "$3" ]]; then
      cp -p "$2" "$1"
    else
      mv "$2" "$1"
    fi
    return 2
  fi
  declare -i LFBFL_is_equal
  if [[ -n "$4" ]]; then
    # Not setting pipefail since the result would still be incorrect.
    # shellcheck disable=SC2312
    diff "$1" "$2"\
      | grep -E "^(>|<)"\
      | grep -v "/$"
    LFBFL_is_equal=${PIPESTATUS[2]}
  else
    diff -q "$1" "$2"
    LFBFL_is_equal=1-$?
  fi
  if [[ LFBFL_is_equal -eq 1 ]]; then
    if [[ -z "$3" ]]; then
      rm "$2"
    fi
    return 0
  fi
  if [[ -n "$3" ]]; then
    cp -p "$2" "$1"
  else
    mv "$2" "$1"
  fi
  return 1
}
```

2.22 build_and_checks_dependencies/python_black_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_bash_option globstar\
    && trap 'enhanced_unset_bash_option globstar' RETURN

  [[ LFBFL_i_verbose -eq 1 ]]\
```

```
    && echo "Checking empty lines after Python function docstrings"
  pcre2grep -M --\
    $'def ([^"]|"(?!"))*"""([^"]|"(?!"))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  check_no_empty_line_after_python_function_docstring "$@"
}
```

2.23 build_and_checks_dependencies/python_isort_complement.libr.sh

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#
```

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  [[ LFBFL_i_verbose -eq 1 ]]\
    && echo "Checking import of _collections_abc is at the right place"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_temp=".check_collections_abc_place.temp"
  local LFBFL_file_name
  find . -name "*.py"\
    | relevant_find\
    | while read -r LFBFL_file_name;
  do
    [[ -f "${LFBFL_file_name}" ]] || continue
    sed -Ez 's/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
      "${LFBFL_file_name}"\
      > "${LFBFL_file_name}${LFBFL_temp}"
    overwrite_if_not_equal "${LFBFL_file_name}"\
      "${LFBFL_file_name}${LFBFL_temp}"
  done
}

python_isort_complement(){
```

```
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  check_collections_abc_place "$@"
}
```

2.24 build_and_checks_dependencies/repository_data.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
```

```

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_name
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_bash_option extglob\
    && trap 'enhanced_unset_bash_option extglob' RETURN

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && echo "You say \"${LFBFL_upgrade_venvs_time_interval_in_seconds}\"."

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) echo "No wat for you?";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && echo "I say \"${LFBFL_upgrade_venvs_time_interval_in_seconds}\"."
}
```

2.25 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php
/*
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```

```
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".
*/

require_once('./build_and_checks_dependencies/split_score.libr.php');

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// The first cast "(int) generate_split_score" is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1], (int) $argv[2], explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
```

2.26 build_and_checks_dependencies/split_score.libr.php

```
<?php
/*
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".
*/

function generate_split_score(
  $b_after_before,
  $i_max_length,
  $a_delimiter_strings_domain,
){
  return function(
    $s_delimiter_string,
    $i_cut_position,
    $b_is_cut_after,
  ) use ($b_after_before, $i_max_length, $a_delimiter_strings_domain){
    if(
      !in_array(
```

```
        $s_delimiter_string, $a_delimiter_strings_domain, true
      )
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_after_before){
      // When $b_after_before,
      // always way better to cut after '/' than before it.
      // When !$b_after_before (aka before_after),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
}
```

2.27 build_and_checks_dependencies/strings_functions.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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# I think I created a file strings_functions.sh or renamed to
```

```
25 # strings_functions.sh at some point, before committing.
26 # Hence, someone must have changed or reversed the name of this file
27 # to string_functions.sh at some point.
28 # This file was renamed from "string_functions.sh" to
29 # "strings_functions.libr.sh".
30
31 LFBFL_subdir="build_and_checks_dependencies"
32 # shellcheck source=get_common_text_glob_patterns.libr.sh
33 source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
34 # shellcheck source=comparisons.libr.sh
35 source "./${LFBFL_subdir}/comparisons.libr.sh"
36
37 split_line_at(){
38   # $1=$line
39   # $2=$position : number of characters in split_line_at_result_beginning
40   declare -g split_line_at_result_beginning="${1:0:$2}"
41   declare -g split_line_at_result_end="${1:$2}"
42   # echo "$split_line_at_result_beginning"
43   # echo "$split_line_at_result_end"
44 }
45
46 declare -gir SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM=1
47 declare -gir SPLIT_SCORE_PROPERTY_POSITION_UNIFORM=2
48 declare -gir SPLIT_SCORE_PROPERTY_LARGER_AFTER=4
49 declare -gir SPLIT_SCORE_PROPERTY_LARGER_BEFORE=8
50
51 split_score_simple(){
52   # $1=$after_before
53   # $2=$delimiters_strings_domain concatenated characters/delimiters
54   # $3=$delimiter_string a single character
55   # $4=$is_cut_after
56   declare -gi split_score_result
57   if [[ "$2" != *$3* ]]; then
58     split_score_result=0
59     return
60   fi
61   if [[ "$4" == "$1" ]]; then
62     split_score_result=2
63     return
64   fi
65   split_score_result=1
66 }
67
68 get_split_score_simple(){
69   # $1=$after_before
70   # $2=$max_length
```

```
  # $3=$delimiters_strings_domain
  declare -g get_split_score_result="split_score_simple $1 '$3'"
  declare -gi get_split_score_result2
  # get_split_score_result2=$((11-$1*4))
  # if [[ $1 -eq 1 ]]; then
  #   get_split_score_result2=7
  # else
  #   get_split_score_result2=11
  # fi
  get_split_score_result2=$((
    SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM
    + SPLIT_SCORE_PROPERTY_POSITION_UNIFORM
    + SPLIT_SCORE_PROPERTY_LARGER_BEFORE
    + (
      $1 * (
        SPLIT_SCORE_PROPERTY_LARGER_AFTER
        - SPLIT_SCORE_PROPERTY_LARGER_BEFORE
      )
    )
  ))
}

get_split_score_exec(){
  declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
  SPLIT_SCORE_EXEC+="/split_score.exec.php"
}

split_score(){
  # $1=$after_before
  # $2=$max_length
  # $3=$delimiters_strings_domain concatenated characters/delimiters
  # $4=$delimiter_string a single character
  # $5=$cut_position
  # $6=$is_cut_after
  get_split_score_exec
  declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
  declare -gi split_score_result
  split_score_result=$(eval "${LFBFL_command}")
}

# This function is currently not called,
# hence we need that shellcheck disable.
# shellcheck disable=SC2034
get_split_score(){
  # $1=$after_before
  # $1=$max_length
```

```
  # $2=$delimiters_strings_domain
  declare -g get_split_score_result="split_score $1 $2 '$3'"
  declare -gi get_split_score_result2
  # get_split_score_result2=$((9-$1*4))
  get_split_score_result2=$((
    SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM
    + SPLIT_SCORE_PROPERTY_LARGER_BEFORE
    + (
      $1 * (
        SPLIT_SCORE_PROPERTY_LARGER_AFTER
        - SPLIT_SCORE_PROPERTY_LARGER_BEFORE
      )
    )
  ))
}

split_line_at_most(){
  # $1=$line
  # $2=$max_length of beginning result string
  # $3=$split_score_command
  #   negative score means do not consider as a possible split
  # $4=$split_score_command_properties :
  #   These properties are flags.
  #   If someday some property is not a flag,
  #   then create split_score_command_properties2.
  #   - null or 0 no property;
  #   - 1 split score doesn't depend on which delimiter matches;
  #   - 2 split score doesn't depend on the current position;
  #   - 4 any split score after is always larger
  #     than any split score before;
  #   - 8 any split score before is always larger
  #     than any split score after;
  # $5=$forbidden_previous_character
  # $6=$split_score_command_delimiter_strings_domain
  #   optimisation if small domain
  #   - this argument can be "null", then we will use $3 only instead,
  #     in that case, only mono-character strings are considered,
  #   - this argument can be an array of characters,
  #     optimize accordingly,
  #   - this argument can be an array of strings,
  #     optimize accordingly,
  #   - this argument can be an array of regexps,
  #     optimize accordingly.
  #     Not sure an array of regexps makes more sense than a single
  #     one. Maybe they could be both cases with corresponding
  #     optimizations. Same for a single character or substring.
```

```
163     #
164     # Note that we're always looking for the last position to split
165     # among maximum score;
166     # if some position gets as score from 2 delimiters,
167     # its score is always the maximum of the two corresponding scores.
168     declare -g split_line_at_most_result_start
169     declare -g split_line_at_most_result_end
170     declare -A LFBFL_positions
171     # echo "$1 $2 $3 $4"
172     # For my use case in bash scripts, I will need only an array of
173     # characters. See get_split_score().
174     if [[ -n "$6" ]]; then
175       echo "split_line_at_most() \$4 NOT IMPLEMENTED YET"
176     fi
177     if [[ $2 -ge ${#1} ]]; then
178       split_line_at_most_result_start=$1
179       split_line_at_most_result_end=""
180       return
181     fi
182     declare -r LFBFL_sort_command="sort --numeric-sort"
183     declare -ir LFBFL_length_minus_1=$(($${#1} - 1))
184     declare -ir LFBFL_i_max=$(
185       min "${LFBFL_sort_command}" "${LFBFL_length_minus_1}" "$2"
186     )
187     declare -i LFBFL_i
188     declare -i LFBFL_j
189     local LFBFL_current_char
190
191     # At the beginning, each position has score 0.
192     for ((LFBFL_i=0; LFBFL_i<=LFBFL_i_max; ++LFBFL_i)) do
193       LFBFL_positions["${LFBFL_i}"]="0"
194     done
195     LFBFL_j=$((LFBFL_i+1))
196     LFBFL_positions["${LFBFL_j}"]="0"
197
198     # If we have a forbidden previous character,
199     # we mark the forbidden positions with -1.
200     if [[ -n "$5" ]]; then
201       for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
202         LFBFL_j=$((LFBFL_i+1))
203         LFBFL_current_char="${1:${LFBFL_i}:1}"
204         if [[ "${LFBFL_current_char}" == "$5" ]]; then
205           LFBFL_positions["${LFBFL_j}"]="-1"
206         fi
207       done
208     fi
```

```
209
210     if [[ "$4" == "7" ]]; then
211       for ((LFBFL_j=LFBFL_i_max; LFBFL_j>0; --LFBFL_j)) do
212         LFBFL_i=$((LFBFL_j-1))
213         LFBFL_current_char="${1:${LFBFL_i}:1}"
214         eval "$3 '${LFBFL_current_char}' 0"
215         # echo "${split_score_result}|${LFBFL_i}"
216         if [[ split_score_result -ge 1 ]]; then
217           # echo "${split_score_result}|${LFBFL_i}"
218           if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
219             LFBFL_positions["${LFBFL_i}"]=$(
220               max "${LFBFL_sort_command}"\
221                 "${LFBFL_positions["${LFBFL_i}"]}"\
222                 "${split_score_result}"
223             )
224           fi
225         fi
226         eval "$3 '${LFBFL_current_char}' 1"
227         # echo "${split_score_result}|${LFBFL_j}"
228         if [[ split_score_result -ge 1 ]]; then
229           # echo "${split_score_result}|${LFBFL_j}"
230           if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
231             LFBFL_positions["${LFBFL_j}"]=$(
232               max "${LFBFL_sort_command}"\
233                 "${LFBFL_positions["${LFBFL_j}"]}"\
234                 "${split_score_result}"
235             )
236             break
237           fi
238         fi
239       done
240     else
241       for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
242         LFBFL_j=$((LFBFL_i+1))
243         LFBFL_current_char="${1:${LFBFL_i}:1}"
244         eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
245         # echo "${split_score_result}|${LFBFL_i}"
246         if [[ split_score_result -ge 1 ]]; then
247           # echo "${split_score_result}|${LFBFL_i}"
248           if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
249             LFBFL_positions["${LFBFL_i}"]=$(
250               max "${LFBFL_sort_command}"\
251                 "${LFBFL_positions["${LFBFL_i}"]}"\
252                 "${split_score_result}"
253             )
254           fi
```

```
255         fi
256         eval "$3 '${LFBFL_current_char}' ${LFBFL_j} 1"
257         # echo "${split_score_result}|${LFBFL_j}"
258         if [[ split_score_result -ge 1 ]]; then
259           # echo "${split_score_result}|${LFBFL_j}"
260           if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
261             LFBFL_positions["${LFBFL_j}"]=$(
262               max "${LFBFL_sort_command}"\
263                 "${LFBFL_positions["${LFBFL_j}"]}"\
264                 "${split_score_result}"
265             )
266           fi
267         fi
268       done
269     fi
270     declare -ir LFBFL_max_score=$(
271       max 'sort --numeric-sort' "${LFBFL_positions[@]}"
272     )
273     local LFBFL_value
274     declare -i LFBFL_best_position
275     for ((LFBFL_i=LFBFL_i_max; LFBFL_i>=0; --LFBFL_i)) do
276       if [[ LFBFL_i -eq 0 ]]; then
277         echo $'/!\\ split_line_at_most bug: no split position found. /!\\'
278       fi
279       local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
280       # echo "${LFBFL_i} ${LFBFL_value}"
281       if [[ "${LFBFL_value}" == "${LFBFL_max_score}" ]]; then
282         LFBFL_best_position=$((LFBFL_i))
283         break
284       fi
285     done
286     # shellcheck disable=SC2248
287     split_line_at "$1" ${LFBFL_best_position}
288     split_line_at_most_result_start="${split_line_at_result_beginning}"
289     split_line_at_most_result_end="${split_line_at_result_end}"
290     # echo "$split_line_at_most_result_start"
291     # echo "$split_line_at_most_result_end"
292   }
293
294   split_last_line(){
295     # $1=$new_lines
296     # $2=$prefix
297     # $3=$max_length
298     # $4=$suffix
299     # $5=$split_score_command
300     # $6=$split_score_command_properties
```

```
  # $7=$forbidden_previous_character ('\' usually)
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
  # This function adapts the effective max position for the split by
  # taking into account the length of the suffix.
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -g split_last_line_result="$1"
  declare -ir LFBFL_max_length_plus=$(($3 + 1))
  declare -ir LFBFL_length2=$(($3 - ${#4}))
  declare -r LFBFL_regexp='.\{'"${LFBFL_max_length_plus}"'\}$'

  # Testing if some line is too long.
  echo "$1"\
    | grep -q "${LFBFL_regexp}";
  if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
    return
  fi

  declare -r LFBFL_start=$(
    echo "$1"\
    | head --lines=-1
  )
  # echo "start: ${LFBFL_start}"
  declare -r LFBFL_last_line=$(
    echo "$1"\
    | tail --lines=1
  )
```

```
347   # echo "last_line: ${LFBFL_last_line}"
348   split_last_line_result=""
349   if [[ -n "${LFBFL_start}" ]]; then
350     split_last_line_result="${LFBFL_start}"
351     # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
352     # echo "${#split_last_line_result}" "${split_last_line_result}"
353     split_last_line_result+=$'\n'
354     # something="${split_last_line_result}"
355     # echo "${#something}" "${something}"
356   fi
357   if [[ -n "$5" ]]; then
358     split_line_at_most "${LFBFL_last_line}" "${LFBFL_length2}"\
359       "$5" "$6" "$7"
360     split_last_line_result+="${split_line_at_most_result_start}"
361     split_last_line_result+="$4"
362     split_last_line_result+=$'\n'
363     split_last_line_result+="$2${split_line_at_most_result_end}"
364   else
365     split_last_line_result+="${LFBFL_last_line:0:${LFBFL_length2}}"
366     split_last_line_result+="$4"
367     split_last_line_result+=$'\n'
368     split_last_line_result+="$2${LFBFL_last_line:${LFBFL_length2}}"
369   fi
370   # echo "result: ${split_last_line_result}"
371 }
372
373 repeated_split_last_line(){
374   # See split_last_line for the first arguments.
375   # $8=final_suffix to handle when the end of the original string
376   #    is the suffix.
377   declare -g repeated_split_last_line_result="$1"
378   local LFBFL_result="$1x"
379   until
380     [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
381   do
382     split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
383       "$4" "$5" "$6" "$7"
384     LFBFL_result="${repeated_split_last_line_result}"
385     repeated_split_last_line_result="${split_last_line_result}"
386     # echo "${repeated_split_last_line_result}"
387   done
388   if [[ -n "$4" && -n "$8" ]]; then
389     declare -ir LFBFL_offset=$((-${#4}))
390     local LFBFL_lines_end
391     LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_offset}}"
392     if [[ "${LFBFL_lines_end}" == "$4" ]]; then
```

```
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

2.28 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

too_long_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
```

```
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  local LFBFL_arg
  declare -i LFBFL_max_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-line-length=}
      LFBFL_max_line_length=$((LFBFL_arg))
      break
    fi
  done

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  local LFBFL_long_line
  local LFBFL_file_name
  local LFBFL_line
  local LFBFL_extension
  local LFBFL_base_name
  declare -ir LFBFL_overlength=$((LFBFL_max_line_length+1))
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 0 ]]\
      || echo "Iterating on pattern: ${LFBFL_pattern}"
    # shellcheck disable=SC2248
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_license_find\
      | xargs grep -H '.\{'${LFBFL_overlength}'\}'\
      | while read -r LFBFL_long_line;
    do
      LFBFL_file_name=${LFBFL_long_line%%:*} # Drop long ':*' suffix
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_name##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_name%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
```

```
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.tpl"
        then
          continue
        fi
        echo "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if ! [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
          echo "${LFBFL_long_line}"
        fi
      else
        echo "${LFBFL_long_line}"
      fi
    done
  done
}
```

2.29 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".
```

```
26
27 source "overwrite_if_not_equal.libr.sh"
28
29 update_common_build_and_checks(){
30   declare -i LFBFL_verbose=0
31   if [[ "$*" == *--verbose* ]]; then
32     echo "$0 $*"
33     LFBFL_verbose=1
34   fi
35   readonly LFBFL_verbose
36
37   if [[ ! -o pipefail ]]; then
38     [[ LFBFL_verbose -eq 1 ]] && echo "pipefail option activated"
39     set -o pipefail
40     trap 'set +o pipefail' RETURN
41   fi
42
43   declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
44   cp "./${LFBFL_common_file_name}.tpl"\
45     "./${LFBFL_common_file_name}.temp"
46
47   declare -ar LFBFL_file_names=(
48     "build_md_from_printable_md.exec.sh"
49     "check_shell_scripts_beginnings.libr.sh"
50     "check_shell_scripts_indentation.libr.sh"
51     "check_URLs.libr.sh"
52     "common_options.libr.sh"
53     "comparisons.libr.sh"
54     "create_PDF.exec.sh"
55     "generate_from_template.libr.sh"
56     "get_common_text_glob_patterns.libr.sh"
57     "grammar_and_spell_check.libr.sh"
58     "licenses_templates/build_licenses_templates.exec.sh"
59     "licenses_templates/license_file_header_AGPLv3+.tpl"
60     "licenses_templates/license_file_header_GPLv3+.tpl"
61     "licenses_templates/license_file_header_LGPLv3+.tpl"
62     "lines_counts.libr.sh"
63     "lines_filters.libr.sh"
64     "lines_maps.libr.sh"
65     "update_or_check_files_names_listing.exec.sh"
66     "overwrite_if_not_equal.libr.sh"
67     "python_black_complement.libr.sh"
68     "python_isort_complement.libr.sh"
69     "repository_data.libr.sh"
70     "split_score.exec.php"
71     "split_score.libr.php"
```

```
    "strings_functions.libr.sh"
    "too_long_code_lines.libr.sh"
  )

  local LFBFL_file_name
  local LFBFL_file_sha512
  local LFBFL_file_sha512_all
  local LFBFL_base_file_name
  for LFBFL_file_name in "${LFBFL_file_names[@]}"; do
    LFBFL_file_sha512=$(
      sha512sum "./${LFBFL_file_name}"\
      | cut -f1 -d' '
    )
    LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    sed -i\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut -f1 -d' '
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed -i\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"
```

```

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

2.30 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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#
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# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
```

```
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  declare -i LFBFL_write=0
  if [[ "$*" == *--write* ]]; then
    LFBFL_write=1
  fi
  readonly LFBFL_write

  declare -i LFBFL_append=0
  if [[ "$*" == *--append* ]]; then
    LFBFL_append=1
  fi
  readonly LFBFL_append

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  LFBFL_subdir2="build_and_checks_variables"
  local LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
  readonly LFBFL_listing
  local LFBFL_data_file_name="./${LFBFL_subdir2}/repository_data.txt"
  readonly LFBFL_data_file_name

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  if [[ LFBFL_write -eq 1 ]]; then
    : > "${LFBFL_listing}"
  fi
  # Remove line returns that are here to keep lines short.
  sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_listing}"\
```

```
    > "${LFBFL_listing}.temp"
  # shopt -s dotglob was needed at some point but I don't see why now.
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_max_line_length} /
  declare -r LFBFL_split_score_command="${get_split_score_result}"
  local LFBFL_split_score_command_properties
  LFBFL_split_score_command_properties="${get_split_score_result2}"
  readonly LFBFL_split_score_command_properties
  local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
  readonly LFBFL_suffix
  # There is always a $'\n' final suffix from >>.
  # But when origin string ends with the suffix we must add something.

  local LFBFL_s_file_paths
  LFBFL_s_file_paths=$(
    find . -type f -printf '%P\n'\
    | relevant_find\
    | sort
  )
  declare -a LFBFL_arr_file_paths
  mapfile -t LFBFL_arr_file_paths <<< "${LFBFL_s_file_paths}"

  declare -r LFBFL_final_suffix=$'\n\\'
  local LFBFL_file_path
  local LFBFL_base_file_name
  for LFBFL_file_path in "${LFBFL_arr_file_paths[@]}"; do
    git check-ignore -q "${LFBFL_file_path}" && continue
    LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
      || continue
    if [[ "${LFBFL_base_file_name}" == *.md ]]; then
      if [[ -f "${LFBFL_file_path}.tpl" ]]; then
        continue
      fi
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      echo "Non-ignored file: ${LFBFL_file_path}"
    fi
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_file_path}"\
      ""\
      ${LFBFL_max_line_length}\
      "${LFBFL_suffix}"\
      "${LFBFL_split_score_command}"\
      "${LFBFL_split_score_command_properties}"\
      ""\
```

```
130         "${LFBFL_final_suffix}"
131       if [[ LFBFL_write -eq 1 ]]; then
132         # shellcheck disable=SC2001
133         echo "${repeated_split_last_line_result}"\
134           >> "${LFBFL_listing}"
135         continue
136       fi
137       if grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
138           "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
139       then
140         :
141       else
142         echo\
143         "The file ${LFBFL_file_path} is not listed in ${LFBFL_listing}."
144         if [[ LFBFL_append -eq 1 ]]; then
145           # shellcheck disable=SC2001
146           echo "${repeated_split_last_line_result}"\
147             >> "${LFBFL_listing}"
148         fi
149       fi
150     done
151
152     LFBFL_s_file_paths=$(cat "${LFBFL_listing}.temp")
153     mapfile -t LFBFL_arr_file_paths <<< "${LFBFL_s_file_paths}"
154
155     for LFBFL_file_path in "${LFBFL_arr_file_paths[@]}"; do
156       [[ "${LFBFL_file_path}" != '//'* ]] || continue
157       LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
158       if ! [[ -f "${LFBFL_file_path}" ]]; then
159         echo\
160         "The non-file ${LFBFL_file_path} is listed in ${LFBFL_listing}."
161       fi
162     done
163
164     rm "${LFBFL_listing}.temp"
165
166     if [[ LFBFL_i_verbose -eq 1 ]]; then
167       cat "${LFBFL_listing}"
168     fi
169   }
170
171   update_or_check_files_names_listing "$@"
```

2.31 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts

```

2.32 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

2.33 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

2.34 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

2.35 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
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```

```
©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
```

```
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
ol.level2{
  counter-reset: level2;
}
ol.level1 li:before{
  content: counter(level1) ". ";
  counter-increment: level1;
}
ol.level2 li:before{
  content: counter(level1) "." counter(level2) " ";
  counter-increment: level2;
}
</style>
<title>@repository_name@</title>
</head>
<body >
<table class="title">
  <tr>
    <td style="padding:1ex">
      <h1 class="titlemain">@repository_name@</h1>
      <h2 class="titlerest">
        @author_full_name@
        <br>
        <a href="@author_website@">
          <span style="font-family:monospace">
          @author_website@
          </span>
        </a>
        <br>
        <a href="mailto:@author_email@">
          <span style="font-family:monospace">
          @author_email@
          </span>
        </a>
      </h2>
    </td>
  </tr>
</table>

<blockquote class="abstract">
```

```
  <span style="font-weight:bold">Abstract:</span>
  @abstract@
</blockquote>
<p>Current version: @current_date@</p>
<p>Current number of commits: @number_of_commits@</p>
<p>Current git SHA1: @current_git_SHA1@</p>
<p>
Code lines: @number_of_lines@
</p>


<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files Tree</a></li>
  <li><a href="#section2">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Listing of files</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
```

```
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

<p>
Bash can be found at the following URLs:<br>
```

```
<a href="https://www.gnu.org/software/bash/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/bash/</span></a>,<br>
<a href="https://tiswww.case.edu/php/chet/bash/bashtop.html"><span
  style="font-family:monospace"
>https://tiswww.case.edu/php/chet/bash/bashtop.html</span></a>,<br>
<a href="https://savannah.gnu.org/projects/bash/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/bash/</span></a>.
</p>

<p>
GNU coreutils can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/coreutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/coreutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/coreutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/coreutils/</span></a>.
</p>

<p>
uutils coreutils can be found at the following URLs:<br>
<a href="https://uutils.github.io/coreutils/"><span
  style="font-family:monospace"
>https://uutils.github.io/coreutils/</span></a>,<br>
<a href="https://github.com/uutils/coreutils"><span
  style="font-family:monospace"
>https://github.com/uutils/coreutils</span></a>.
</p>

<p>
diff (GNU diffutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/diffutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/diffutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/diffutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/diffutils/</span></a>.
</p>

<p>
find, xargs (GNU findutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/findutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/findutils/</span></a>,<br>
```

```
<a href="https://savannah.gnu.org/projects/findutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/findutils/</span></a>.
</p>

<p>
grep (GNU grep) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/grep/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/grep/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/grep/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/grep/</span></a>.
</p>

<p>
sed (GNU sed) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/sed/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/sed/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/sed/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/sed/</span></a>.
</p>

<p>
ShellCheck can be found at the following URLs:<br>
<a href="https://www.shellcheck.net/"><span
  style="font-family:monospace"
>https://www.shellcheck.net/</span></a>,<br>
<a href="https://github.com/koalaman/shellcheck"><span
  style="font-family:monospace"
>https://github.com/koalaman/shellcheck</span></a>.
</p>

<p>
GNU Wget can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/wget/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/wget/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/wget/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/wget/</span></a>.
</p>

<p>
```

```
more of util-linux can be found at the following URLs:<br>
<a href="https://www.kernel.org/pub/linux/utils/util-linux/"><span
  style="font-family:monospace"
>https://www.kernel.org/pub/linux/utils/util-linux/</span></a>,<br>
<a
href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
><span
  style="font-family:monospace"
>https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
></a>,<br>
<a href="https://github.com/util-linux/util-linux"><span
  style="font-family:monospace"
>https://github.com/util-linux/util-linux</span></a> official?
</p>

<p>
pandoc can be found at the following URLs:<br>
<a href="https://pandoc.org/"><span
  style="font-family:monospace"
>https://pandoc.org/</span></a>,<br>
<a href="https://github.com/jgm/pandoc"><span
  style="font-family:monospace"
>https://github.com/jgm/pandoc</span></a>.
</p>

<p>
pdftex/pdflatex can be found at the following URLs:<br>
<a href="https://www.tug.org/applications/pdftex/"><span
  style="font-family:monospace"
>https://www.tug.org/applications/pdftex/</span></a>,<br>
<a href="https://tug.org/texlive/svn/"><span
  style="font-family:monospace"
>https://tug.org/texlive/svn/</span></a>.
</p>

<p>
tree can be found at the following URLs:<br>
<a href="https://oldmanprogrammer.net/source.php?dir=projects/tree"
><span style="font-family:monospace"
>https://oldmanprogrammer.net/source.php?dir=projects/tree</span></a>,<br>
<a href="https://gitlab.com/OldManProgrammer/unix-tree"><span
  style="font-family:monospace"
>https://gitlab.com/OldManProgrammer/unix-tree</span></a>,<br>
<a href="https://github.com/Old-Man-Programmer/tree"><span
  style="font-family:monospace"
>https://github.com/Old-Man-Programmer/tree</span></a>.
```

```
</p>

<p>
git can be found at the following URLs:<br>
<a href="https://git-scm.com/about"><span
  style="font-family:monospace"
>https://git-scm.com/about</span></a>,<br>
<a href="https://github.com/git/git"><span
  style="font-family:monospace"
>https://github.com/git/git</span></a>.
</p>

<p>
pylint can be found at the following URLs:<br>
<a href="https://www.pylint.org/"><span
  style="font-family:monospace"
>https://www.pylint.org/</span></a>,<br>
<a href="https://pylint.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://pylint.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/pylint-dev/pylint"><span
  style="font-family:monospace"
>https://github.com/pylint-dev/pylint</span></a>,<br>
<a href="https://pypi.org/project/pylint/"><span
  style="font-family:monospace"
>https://pypi.org/project/pylint/</span></a>.
</p>

<p>
notify-send can be found at the following URLs:<br>
<a href="https://gitlab.gnome.org/GNOME/libnotify/"><span
  style="font-family:monospace"
>https://gitlab.gnome.org/GNOME/libnotify/</span></a>,<br>
<a href="https://gnome.pages.gitlab.gnome.org/libnotify/"><span
  style="font-family:monospace"
>https://gnome.pages.gitlab.gnome.org/libnotify/</span></a>.
</p>

<p>
black can be found at the following URLs:<br>
<a href="https://github.com/psf/black"><span
  style="font-family:monospace"
>https://github.com/psf/black</span></a>,<br>
<a href="https://black.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://black.readthedocs.io/en/stable/</span></a>,<br>
```

```
<a href="https://pypi.org/project/black/"><span
  style="font-family:monospace"
>https://pypi.org/project/black/</span></a>.
</p>

<p>
isort can be found at the following URLs:<br>
<a href="https://pycqa.github.io/isort/"><span
  style="font-family:monospace"
>https://pycqa.github.io/isort/</span></a>,<br>
<a href="https://github.com/pycqa/isort/"><span
  style="font-family:monospace"
>https://github.com/pycqa/isort/</span></a>,<br>
<a href="https://pypi.org/project/isort/"><span
  style="font-family:monospace"
>https://pypi.org/project/isort/</span></a>.
</p>

<p>
mypy can be found at the following URLs:<br>
<a href="https://mypy-lang.org/"><span
  style="font-family:monospace"
>https://mypy-lang.org/</span></a>,<br>
<a href="https://mypy.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://mypy.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/python/mypy"><span
  style="font-family:monospace"
>https://github.com/python/mypy</span></a>,<br>
<a href="https://pypi.org/project/mypy/"><span
  style="font-family:monospace"
>https://pypi.org/project/mypy/</span></a>.
</p>

<p>
ruff can be found at the following URLs:<br>
<a href="https://docs.astral.sh/ruff/"><span
  style="font-family:monospace"
>https://docs.astral.sh/ruff/</span></a>,<br>
<a href="https://github.com/astral-sh/ruff"><span
  style="font-family:monospace"
>https://github.com/astral-sh/ruff</span></a>,<br>
<a href="https://pypi.org/project/ruff/"><span
  style="font-family:monospace"
>https://pypi.org/project/ruff/</span></a>.
</p>
```

```
<p>
bandit can be found at the following URLs:<br>
<a href="https://bandit.readthedocs.io/en/latest/"><span
  style="font-family:monospace"
>https://bandit.readthedocs.io/en/latest/</span></a>,<br>
<a href="https://github.com/PyCQA/bandit"><span
  style="font-family:monospace"
>https://github.com/PyCQA/bandit</span></a>,<br>
<a href="https://pypi.org/project/bandit/"><span
  style="font-family:monospace"
>https://pypi.org/project/bandit/</span></a>.
</p>

<p>
phpmd can be found at the following URLs:<br>
<a href="https://phpmd.org/"><span
  style="font-family:monospace"
>https://phpmd.org/</span></a>,<br>
<a href="https://github.com/phpmd/phpmd"><span
  style="font-family:monospace"
>https://github.com/phpmd/phpmd</span></a>,<br>
<a href="https://packagist.org/packages/phpmd/phpmd"><span
  style="font-family:monospace"
>https://packagist.org/packages/phpmd/phpmd</span></a>.
</p>

<p>
php can be found at the following URLs:<br>
<a href="https://www.php.net/"><span
  style="font-family:monospace"
>https://www.php.net/</span></a>,<br>
<a href="https://github.com/php/php-src"><span
  style="font-family:monospace"
>https://github.com/php/php-src</span></a>.
</p>

<p>
tex can be found at the following URL:<br>
<a href="https://ctan.org/tex-archive/systems/knuth/dist/tex"><span
  style="font-family:monospace"
>https://ctan.org/tex-archive/systems/knuth/dist/tex</span></a>.
</p>

<p>
pcregrep, pcre2grep can be found at the following URLs:<br>
```

```
<a href="https://www.pcre.org/"><span
  style="font-family:monospace"
>https://www.pcre.org/</span></a>,<br>
<a href="https://pcre2project.github.io/pcre2/"><span
  style="font-family:monospace"
>https://pcre2project.github.io/pcre2/</span></a>,<br>
<a href="https://github.com/PCRE2Project/pcre1"><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre1</span></a>,<br>
<a href="https://github.com/PCRE2Project/pcre2"><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre2</span></a>,<br>
<a
href="https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c</span
></a>.
</p>

<p>
If you see something that I missed regarding my dependencies,
please tell me/email me :).
</p>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
```

```
);
</script>
</html>
```

2.36 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
```

```
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

```

```
\section{Files tree}
\label{section:tree}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Listing of files}
\label{section:listing}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
```

```
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:\newline
\url{https://www.gnu.org/software/bash/},\newline
\url{https://tiswww.case.edu/php/chet/bash/bashtop.html},\newline
\url{https://savannah.gnu.org/projects/bash/}.

GNU coreutils can be found at the following URLs:\newline
\url{https://www.gnu.org/software/coreutils/},\newline
\url{https://savannah.gnu.org/projects/coreutils/}.

uutils coreutils can be found at the following URLs:\newline
\url{https://uutils.github.io/coreutils/},\newline
\url{https://github.com/uutils/coreutils}.

diff (GNU diffutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/diffutils/},\newline
\url{https://savannah.gnu.org/projects/diffutils/}.

find, xargs (GNU findutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/findutils/},\newline
\url{https://savannah.gnu.org/projects/findutils/}.

grep (GNU grep) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/grep/},\newline
\url{https://savannah.gnu.org/projects/grep/}.

sed (GNU sed) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/sed/},\newline
\url{https://savannah.gnu.org/projects/sed/}.

ShellCheck can be found at the following URLs:\newline
\url{https://www.shellcheck.net/},\newline
\url{https://github.com/koalaman/shellcheck}.

GNU Wget can be found at the following URLs:\newline
\url{https://www.gnu.org/software/wget/},\newline
\url{https://savannah.gnu.org/projects/wget/}.
```

```
more of util-linux can be found at the following URLs:\newline
\url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
{\small
  \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
},
\newline
\url{https://github.com/util-linux/util-linux} official?

pandoc can be found at the following URLs:\newline
\url{https://pandoc.org/},\newline
\url{https://github.com/jgm/pandoc}.

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\url{https://www.tug.org/applications/pdftex/},\newline
\url{https://tug.org/texlive/svn/}.

tree can be found at the following URLs:\newline
\url{https://oldmanprogrammer.net/source.php?dir=projects/tree},\newline
\url{https://gitlab.com/OldManProgrammer/unix-tree},\newline
\url{https://github.com/Old-Man-Programmer/tree}.

git can be found at the following URLs:\newline
\url{https://git-scm.com/about},\newline
\url{https://github.com/git/git}.

pylint can be found at the following URLs:\newline
\url{https://www.pylint.org/},\newline
\url{https://pylint.readthedocs.io/en/stable/},\newline
\url{https://github.com/pylint-dev/pylint},\newline
\url{https://pypi.org/project/pylint/}.

notify-send can be found at the following URLs:\newline
\url{https://gitlab.gnome.org/GNOME/libnotify/},\newline
\url{https://gnome.pages.gitlab.gnome.org/libnotify/}.

black can be found at the following URLs:\newline
\url{https://github.com/psf/black},\newline
\url{https://black.readthedocs.io/en/stable/},\newline
\url{https://pypi.org/project/black/}.

isort can be found at the following URLs:\newline
\url{https://pycqa.github.io/isort/},\newline
\url{https://github.com/pycqa/isort/},\newline
\url{https://pypi.org/project/isort/}.

mypy can be found at the following URLs:\newline
```

```
\url{https://mypy-lang.org/},\newline
\url{https://mypy.readthedocs.io/en/stable/},\newline
\url{https://github.com/python/mypy},\newline
\url{https://pypi.org/project/mypy/}.

ruff can be found at the following URLs:\newline
\url{https://docs.astral.sh/ruff/},\newline
\url{https://github.com/astral-sh/ruff},\newline
\url{https://pypi.org/project/ruff/}.

bandit can be found at the following URLs:\newline
\url{https://bandit.readthedocs.io/en/latest/},\newline
\url{https://github.com/PyCQA/bandit},\newline
\url{https://pypi.org/project/bandit/}.

phpmd can be found at the following URLs:\newline
\url{https://phpmd.org/},\newline
\url{https://github.com/phpmd/phpmd},\newline
\url{https://packagist.org/packages/phpmd/phpmd}.

php can be found at the following URLs:\newline
\url{https://www.php.net/},\newline
\url{https://github.com/php/php-src}.

tex can be found at the following URL:\newline
\url{https://ctan.org/tex-archive/systems/knuth/dist/tex}.

pcregrep, pcre2grep can be found at the following URLs:\newline
\url{https://www.pcre.org/},\newline
\url{https://pcre2project.github.io/pcre2/},\newline
\url{https://github.com/PCRE2Project/pcre1},\newline
\url{https://github.com/PCRE2Project/pcre2},\newline
{\small
  \url{https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c}
}.

If you see something that I missed regarding my dependencies,
please tell me/email me :).

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

2.37 build_and_checks_variables/files_listing.tex.tpl

```
\subsection{
  build\_and\_checks\_dependencies/build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/build\_md\_from\_printable\_md.exec.sh
}
\label{
  build_and_checks_dependencies:build_md_from_printable_mdexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_beginnings.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_beginningslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_indentation.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_indentationlibrsh
}
```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/check\_URLs.libr.sh
}
\label{
  build_and_checks_dependencies:check_URLslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_URLs.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/common\_options.libr.sh
}
\label{
```

```
  build_and_checks_dependencies:common_optionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_options.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/comparisons.libr.sh
}
\label{
  build_and_checks_dependencies:comparisonslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/comparisons.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/create\_PDF.exec.sh
}
\label{
  build_and_checks_dependencies:create_PDFexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/create_PDF.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/generate\_from\_template.libr.sh
}
\label{
  build_and_checks_dependencies:generate_from_templatelibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/generate_from_template.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
```

```
get\_common\_text\_glob\_patterns.libr.sh
}
\label{
  build_and_checks_dependencies:get_common_text_glob_patternslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/grammar\_and\_spell\_check.libr.sh
}
\label{
  build_and_checks_dependencies:grammar_and_spell_checklibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
build\_licenses\_templates.exec.sh
}
\label{
  build_and_checks_dependencies:licenses_templates:%
build_licenses_templatesexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
build_licenses_templates.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_AGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_AGPLv3+tpl
}
```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_AGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_GPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_GPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_GPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_LGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_LGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_LGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/lines\_counts.libr.sh
}
\label{
  build_and_checks_dependencies:lines_countslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_counts.libr.sh
```

```
}


\subsection{
  build\_and\_checks\_dependencies/lines\_filters.libr.sh
}
\label{
  build_and_checks_dependencies:lines_filterslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_filters.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_maps.libr.sh
}
\label{
  build_and_checks_dependencies:lines_mapslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_maps.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/overwrite\_if\_not\_equal.libr.sh
}
\label{
  build_and_checks_dependencies:overwrite_if_not_equallibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_black\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_black_complementlibrsh
}

```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_black_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_isort\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_isort_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_isort_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/repository\_data.libr.sh
}
\label{
  build_and_checks_dependencies:repository_datalibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/repository_data.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.exec.php
}
\label{
  build_and_checks_dependencies:split_scoreexecphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.exec.php
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.libr.php
}
\label{
  build_and_checks_dependencies:split_scorelibrphp
```

```
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.libr.php
}


\subsection{
  build\_and\_checks\_dependencies/strings\_functions.libr.sh
}
\label{
  build_and_checks_dependencies:strings_functionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/strings_functions.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/too\_long\_code\_lines.libr.sh
}
\label{
  build_and_checks_dependencies:too_long_code_lineslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/too_long_code_lines.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:update_common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
```

```
update\_or\_check\_files\_names\_listing.exec.sh
}
\label{
  build_and_checks_dependencies:update_or_check_files_names_listingexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh
}


\subsection{
  build\_and\_checks\_variables/.gitignore
}
\label{
  build_and_checks_variables:gitignore
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/.gitignore
}


\subsection{
  build\_and\_checks\_variables/bandit.ini
}
\label{
  build_and_checks_variables:banditini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit.ini
}


\subsection{
  build\_and\_checks\_variables/bandit\_baseline.json
}
\label{
  build_and_checks_variables:bandit_baselinejson
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit_baseline.json
}

```

```
\subsection{
  build\_and\_checks\_variables/black.toml
}
\label{
  build_and_checks_variables:blacktoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/black.toml
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.html.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptshtmltpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.tex.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptstextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_listing.tex.tpl
}
\label{
  build_and_checks_variables:files_listingtextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_listing.tex.tpl
```

```
}


\subsection{
  build\_and\_checks\_variables/files\_names\_listing.txt
}
\label{
  build_and_checks_variables:files_names_listingtxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_names_listing.txt
}


\subsection{
  build\_and\_checks\_variables/licenses\_notes.txt
}
\label{
  build_and_checks_variables:licenses_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/licenses_notes.txt
}


\subsection{
  build\_and\_checks\_variables/phpmd\_baseline.xml
}
\label{
  build_and_checks_variables:phpmd_baselinexml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/phpmd_baseline.xml
}


\subsection{
  build\_and\_checks\_variables/pylintrc.toml
}
\label{
  build_and_checks_variables:pylintrctoml
}

```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/pylintrc.toml
}


\subsection{
  build\_and\_checks\_variables/repository\_data.txt
}
\label{
  build_and_checks_variables:repository_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/repository_data.txt
}


\subsection{
  build\_and\_checks\_variables/ruff.toml
}
\label{
  build_and_checks_variables:rufftoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/ruff.toml
}


\subsection{
  build\_and\_checks\_variables/shellcheck.ini
}
\label{
  build_and_checks_variables:shellcheckini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck.ini
}


\subsection{
  build\_and\_checks\_variables/shellcheck\_notes.txt
}
\label{
  build_and_checks_variables:shellcheck_notestxt
```

```
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck_notes.txt
}


\subsection{
  build\_and\_checks.exec.sh
}
\label{
  build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks.exec.sh
}


\subsection{
  COPYING
}
\label{
  COPYING
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING
}


\subsection{
  COPYING.LESSER
}
\label{
  COPYINGLESSER
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING.LESSER
}


\subsection{
  git\_helpers/git\_checks.libr.sh
}
```

```
\label{
  git_helpers:git_checkslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/git_checks.libr.sh
}


\subsection{
  git\_helpers/post-commit
}
\label{
  git_helpers:post-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post-commit
}


\subsection{
  git\_helpers/post\_commit\_renaming\_comment.libr.sh
}
\label{
  git_helpers:post_commit_renaming_commentlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post_commit_renaming_comment.libr.sh
}


\subsection{
  git\_helpers/pre-commit
}
\label{
  git_helpers:pre-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/pre-commit
}


\subsection{
```

```
  IMAP\_transfert.py
}
\label{
  IMAP_transfertpy
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
IMAP_transfert.py
}


\subsection{
  LiensUtiles.txt
}
\label{
  LiensUtilestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
LiensUtiles.txt
}


\subsection{
  miscellaneous/funny.sh
}
\label{
  miscellaneous:funnysh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/funny.sh
}


\subsection{
  miscellaneous/spampty\_data.txt
}
\label{
  miscellaneous:spampty_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/spampty_data.txt
}

```

```

\subsection{
  miscellaneous/%
yet\_another\_long\_long\_long\_long\_long\_long\_long\_long\_long\_filena%
me\_so\_that\_you\_can\_see\_that\_it\_works.txt
}
\label{
  miscellaneous:%
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_%
you_can_see_that_it_workstxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/%
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_%
you_can_see_that_it_works.txt
}


\subsection{
  README.md.tpl
}
\label{
  READMEmdtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
README.md.tpl
}


\subsection{
  unicode.libr.php
}
\label{
  unicodelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
unicode.libr.php
}


\subsection{
  upgrade\_build\_and\_checks.exec.sh
}
```

```
\label{
  upgrade_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
upgrade_build_and_checks.exec.sh
}


\subsection{
  wget\_sha512.libr.sh
}
\label{
  wget_sha512librsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
wget_sha512.libr.sh
}
```

2.38 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
```

```
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
```

```
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_listing.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234567890123456789012345
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901234\
56
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901234\
56\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901234\
5\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901234\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901234567890123\
```

```
\
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/spampty_data.txt
miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

2.39 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception.

Remark: Some commands listed under GPLv3+ are POSIX
```

```
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

```

```
pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
```

```
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

2.40 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName" file="build_and
  <violation rule="PHPMD\Rule\Naming\LongVariable" file="build_and_checks_dependenc
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName" file="build_and_
  <violation rule="PHPMD\Rule\Design\LongMethod" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName" file="unicode.l
  <violation rule="PHPMD\Rule\Naming\LongVariable" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName" file="unicode.li
  <violation rule="PHPMD\Rule\CleanCode\MissingImport" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\CyclomaticComplexity" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Design\NpathComplexity" file="unicode.libr.php"/>
</phpmd-baseline>
```

2.41 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

2.42 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
```

```
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

grammar_or_spell_checker_command=more


Why one venv for each Python dependency ?
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
```

```
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
        }
      },
      "D": {
        "0.1.0": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "4.4.4" : {}
          }
        }
      }
    }
  },
  "B": {
    "2.2.2": {
      "C": {
        // Dumb, you did not found it by yourself? Failed exam.
        "4.4.4" : {}
      }
    }
  }
}
Wonderful someone rediscovered nested structures XD
Seriously so many engineers or computer scientists are dumb to cry.

bandit_venv=../venv_bandit
black_venv=../venv_black
isort_venv=../venv_isort
mypy_venv=../venv_mypy
pylint_venv=../venv_pylint
ruff_venv=../venv_ruff


upgrade venvs can be auto or ask, any other value is interpreted
as ask.

upgrade_venvs=ask
```

```


time interval can be 0 to always upgrade or ask to upgrade,
or a number at most bash max int
https://superuser.com/questions/1030122/
what-is-the-maximum-value-of-a-numeric-bash-shell-variable
it should be enough to simulate never "upgrade or ask for upgrade"
unless you claim to live many billions years ;)
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800
```

2.43 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

```

```
  # commented-out-code Of course git log is always faster...
  # Note that if you’re tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

2.44 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"

```

2.45 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck:
- SC1090 : Can’t follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
```

```
            Only remaining cases are in common_build_and_checks
            when sourcing some Python venv
            source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
            False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
            False positive when c contains \n.
- SC2034 : Unused variable
            False positive for globals that are declared in functions
            that are not currently called.
            Somehow it can be used to detect dead code.
- SC2086 : Double quote to prevent globbing and word splitting.
            False positive when returning integer variable
            via an indirection like return ${!foo}.
            See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
            False positive when passing arguments would yield
            a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
            False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
            False positive when assigning a variable declared with -i.
            https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
            False positive when using "$@" line return and pipes.
            https://github.com/koalaman/shellcheck/issues/3192
            Example:
              # shellcheck disable=SC2145
              all_self_code_lines "$@"\
                | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
            Very useful but false positive when I play with
            variable variable names.
- SC2162 : read without -r will mangle backslashes.
            False positive when read is used without variable:
            https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
            False positive on returning integer values.
            https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
              Use = to compare as string (or expand explicitly with $var).
            Very useful but false positive when I play with
            variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
            value is masked. Frequent with pipes.

```

```
The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2034$" -e "=SC2086$" -e "=SC2119$" -e "=SC2120$" -e "=SC2125$"\
  -e "=SC2145$" -e "=SC2154$" -e "=SC2162$" -e "=SC2248$" -e "=SC2309$"\
  -e "=SC2312$" -e "=SC2154,SC2309$" -e "=SC2162,SC2312$"
```

2.46 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

```

```
38    local LFBFL_verbose=""
39    if [[ "$*" == *--verbose* ]]; then
40      echo "$0 $*"
41      LFBFL_verbose="--verbose"
42    fi
43    readonly LFBFL_verbose
44
45    source ./wget_sha512.libr.sh
46
47    LFBFL_subdir="build_and_checks_dependencies"
48    mkdir -p "${LFBFL_subdir}/licenses_templates"
49    LFBFL_variables_directory="${LFBFL_work_directory}/"
50    LFBFL_variables_directory+="build_and_checks_variables"
51    mkdir -p "${LFBFL_variables_directory}/temp"
52    if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
53      {
54        echo "temp/"
55        echo ""
56      } > "${LFBFL_variables_directory}/.gitignore"
57    fi
58
59    # LFBFL_dependencies_raw_content_download_URL
60    local LFBFL_dependencies_URL
61    LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
62    LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
63    readonly LFBFL_dependencies_URL
64
65    declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
66    # LFBFL_script_download_URL
67    declare -r\
68      LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
69    declare -r\
70      LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
71    local LFBFL_correct_sha512
72    LFBFL_correct_sha512='2a67f603bba1c301a237b6cba57a7aee38994f5231ffb'
73    LFBFL_correct_sha512+='8d4e50667eea63f89181e02e1cb352122b21ed7b2711'
74    LFBFL_correct_sha512+='d13a6ef2ef09bbfe9db759cb8455b18b5d36bcc'
75    wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
76      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
77    chmod +x "./${LFBFL_file_path}"
78
79    "${LFBFL_file_path}" "${LFBFL_work_directory}"\
80      "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
81  }
82
83  if [[ "$*" == *--fixed_point_build* ]]; then
```

```
  echo "--fixed_point_build"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed_point_build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.47 COPYING

```
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                       Version 3, 29 June 2007

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```
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```

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```
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```

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```
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```

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```
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        Library.  A suitable mechanism is one that (a) uses at run time
        a copy of the Library already present on the user's computer
        system, and (b) will operate properly with a modified version
        of the Library that is interface-compatible with the Linked
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    e) Provide Installation Information, but only if you would otherwise
    be required to provide such information under section 6 of the
    GNU GPL, and only to the extent that such information is
    necessary to install and execute a modified version of the
```

```
    Combined Work produced by recombining or relinking the
    Application with a modified version of the Linked Version. (If
    you use option 4d0, the Installation Information must accompany
    the Minimal Corresponding Source and Corresponding Application
    Code. If you use option 4d1, you must provide the Installation
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    on the Library, uncombined with any other library facilities,
    conveyed under the terms of this License.

    b) Give prominent notice with the combined library that part of it
    is a work based on the Library, and explaining where to find the
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```

2.49 git_helpers/git_checks.libr.sh

```
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# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
```

```
45    declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
46    declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
47    local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
48    LFBFL_send_body_2+=" dès la création du modèle"
49    readonly LFBFL_send_body_2
50    declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
51    declare -r LFBFL_send_body_3=\
52  "Les nouveaux preloadings sont interdits -> #Prefetch()"
53    declare -i LFBFL_error=0
54    git diff --cached --name-only\
55      > /tmp/DOSAS_django_git_check1.temp
56    if grep models /tmp/DOSAS_django_git_check1.temp;
57    then
58      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
59      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
60      LFBFL_error=1
61    fi
62    if grep serializer /tmp/DOSAS_django_git_check1.temp;
63    then
64      notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
65      LFBFL_error=1
66    fi
67    rm /tmp/DOSAS_django_git_check1.temp
68    # shellcheck disable=SC2248
69    return ${LFBFL_error}
70  }
71
72
73  check_no_abusive_trailing_comma(){
74    # Useful for Python code, although it regularly yields
75    # false positives in Python code
76    # (but it is easy to distinguish by hand).
77    # Use `git commit --no-verify` after check.
78    declare -r LFBFL_send_summary_1="ATTENTION"
79    local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
80    LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
81    readonly LFBFL_send_body_1
82    declare -i LFBFL_error=0
83    git diff --cached -r\
84      > /tmp/DOSAS_django_git_check2.temp
85    if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
86    then
87      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
88      LFBFL_error=1
89    fi
90    rm /tmp/DOSAS_django_git_check2.temp
```

```
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_files_string=$(
    git diff --cached --name-only\
    | grep '\.py'
  )
  echo "${LFBFL_files_string}"
  declare -a LFBFL_some_files
  mapfile -t LFBFL_some_files <<< "${LFBFL_files_string}"
  readonly LFBFL_some_files
  declare -i LFBFL_error=0
  # Not very useful since LFBFL_file should be local to the loop by
  # default.
  local LFBFL_file
  for LFBFL_file in "${LFBFL_some_files[@]}"; do
    echo "Black will check formatting on file ${LFBFL_file}"
    if ! black --check --diff "${LFBFL_file}";
    then
      LFBFL_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}
```

2.50 git_helpers/post-commit

```
#!/usr/bin/env bash
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```

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#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_library="../DevOrSysAdminScripts/git_helpers/"
LFBFL_library+="post_commit_renaming_comment.libr.sh"
# shellcheck source=post_commit_renaming_comment.libr.sh
source "${LFBFL_library}"
LFBFL_library="../DevOrSysAdminScripts/build_and_checks_dependencies/"
LFBFL_library+="lines_filters.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "${LFBFL_library}"

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

2.51 git_helpers/post_commit_renaming_comment.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_cd_result2
pushd .
cd "${LFBFL_some_directory}" || {
  LFBFL_cd_result2=$?
  echo "post_commit_renaming_comment.libr.sh no such directory"
  # shellcheck disable=SC2248
  exit ${LFBFL_cd_result2}
}
# shellcheck source=common_options.libr.sh
source "build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
```

```
50 source "build_and_checks_dependencies/strings_functions.libr.sh"
51 declare -i LFBFL_popd_result2
52 popd || {
53   LFBFL_popd_result2=$?
54   echo "post_commit_renaming_comment.libr.sh no popd"
55   # shellcheck disable=SC2248
56   exit ${LFBFL_popd_result2}
57 }
58
59 # Never forget that the worse security is the illusion of security
60 # (Kevin Mitnick). So, do not expect to have an accurate trackability
61 # of code history just because you use this post-commit hook or
62 # something else more evolved. This is useful but not a guarantee.
63 # For this reason, this function will only handle when all files
64 # impacted in the preceding commit are only renamed.
65 # It assumes that `diff` is the current default difference utility
66 # parametered with `git`. This point could be enhanced.
67 commit_a_file_renamed_comment(){
68   # Options:
69   #   --verbose
70   #   --log-directory-change
71   #   --max-comment-line-length
72   declare -i LFBFL_i_verbose=0
73   get_verbose_option "$@"
74   local LFBFL_work_directory=""
75   get_work_directory_option "$@"
76   local LFBFL_arg
77
78   declare -i LFBFL_log_directory_change=0
79   if [[ "$*" == *--log-directory-change* ]]; then
80     LFBFL_log_directory_change=1
81   fi
82   readonly LFBFL_log_directory_change
83
84   declare -i LFBFL_max_comment_line_length=70
85   for LFBFL_arg in "$@"; do
86     if [[ "${LFBFL_arg}" == --max-comment-line-length=* ]]; then
87       LFBFL_arg=${LFBFL_arg#--max-comment-line-length=}
88       LFBFL_max_comment_line_length=$((LFBFL_arg))
89       break
90     fi
91   done
92   readonly LFBFL_max_comment_line_length
93
94   if [[ LFBFL_i_verbose -eq 1 ]]; then
95     echo "LFBFL_log_directory_change: ${LFBFL_log_directory_change}"
```

```
96      echo "LFBFL_max_comment_line_length:"\
97        "${LFBFL_max_comment_line_length}"
98    fi
99    #-----------------------------------------------------------------
100
101   enhanced_set_shell_option pipefail\
102     && trap 'enhanced_unset_shell_option pipefail' RETURN
103
104   declare -r LFBFL_renaming_lines=$(
105     git log -p -1\
106     | grep '^diff --git' -A 1
107   )
108   if [[ LFBFL_i_verbose -eq 1 ]]; then
109     echo "LFBFL_renaming_lines: ${LFBFL_renaming_lines}"
110   fi
111
112   declare -a LFBFL_renaming_lines_array
113   mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
114   readonly LFBFL_renaming_lines_array
115
116   declare -r LFBFL_timestamp=$(date --iso-8601="minute")
117   if [[ LFBFL_i_verbose -eq 1 ]]; then
118     echo "LFBFL_timestamp: ${LFBFL_timestamp}"
119   fi
120   get_split_score_simple 1 "${LFBFL_max_comment_line_length}" _
121   LFBFL_split_score_command1="${get_split_score_result}"
122   LFBFL_split_score_command_properties1="${get_split_score_result2}"
123   get_split_score_simple 1 "${LFBFL_max_comment_line_length}" /
124   LFBFL_split_score_command2="${get_split_score_result}"
125   LFBFL_split_score_command_properties2="${get_split_score_result2}"
126   declare -i LFBFL_renaming_happened=0
127   declare -i LFBFL_i
128   declare -i LFBFL_j
129   # declare -r LFBFL_special_file_name="files_names_listing.txt"
130   declare -r LFBFL_sfn="files_names_listing.txt"
131   local LFBFL_diff_line
132   local LFBFL_similarity_line
133   local LFBFL_old_file_path
134   local LFBFL_old_file_name
135   local LFBFL_old_file_directory
136   local LFBFL_new_file_path
137   local LFBFL_new_file_name
138   local LFBFL_new_file_directory
139   local LFBFL_useful_file_name
140   local LFBFL_extension
141   local LFBFL_comment_prefix
```

```
142   local LFBFL_comment_prefix2
143   local LFBFL_old_file_name2
144   local LFBFL_old_file_directory2
145   local LFBFL_new_file_name2
146   local LFBFL_new_file_directory2
147   local LFBFL_new_comment
148   declare -i LFBFL_copyright_line_number
149   declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
150   declare -i LFBFL_line_count
151   declare -i LFBFL_lines_after
152   local LFBFL_temp_file_path
153   for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
154     LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
155     LFBFL_i=$((LFBFL_i + 1))
156     LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
157     LFBFL_i=$((LFBFL_i + 1))
158     # We need to increment 3 times since grep adds "--" lines between.
159     LFBFL_i=$((LFBFL_i + 1))
160     if [[ LFBFL_i_verbose -eq 1 ]]; then
161       echo "LFBFL_diff_line: ${LFBFL_diff_line}"
162       echo "LFBFL_similarity_line: ${LFBFL_similarity_line}"
163     fi
164     if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
165       continue
166     fi
167
168     # Extract old file name from diff line. --------------------------
169     LFBFL_old_file_path=$(
170       echo "${LFBFL_diff_line}"\
171       | cut -d ' ' -f 3
172     )
173     # It starts by "a/".
174     LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
175     LFBFL_old_file_name=$(basename "${LFBFL_old_file_path}")
176     LFBFL_old_file_directory=$(dirname "${LFBFL_old_file_path}")
177     # Extract new file name from diff line. --------------------------
178     LFBFL_new_file_path=$(
179       echo "${LFBFL_diff_line}"\
180       | cut -d ' ' -f 4
181     )
182     # It starts by "b/".
183     LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
184     LFBFL_new_file_name=$(basename "${LFBFL_new_file_path}")
185     LFBFL_new_file_directory=$(dirname "${LFBFL_new_file_path}")
186     if [[ LFBFL_i_verbose -eq 1 ]]; then
187       echo "LFBFL_old_file_path: ${LFBFL_old_file_path}"
```

```
188         echo "LFBFL_old_file_name: ${LFBFL_old_file_name}"
189         echo "LFBFL_old_file_directory: ${LFBFL_old_file_directory}"
190         echo "LFBFL_new_file_path: ${LFBFL_new_file_path}"
191         echo "LFBFL_new_file_name: ${LFBFL_new_file_name}"
192         echo "LFBFL_new_file_directory: ${LFBFL_new_file_directory}"
193       fi
194       # Extract "LFBFL_extension" with smart guessing for .tpl ---------
195       # and set prefix for line with ©Copyright. -----------------------
196       LFBFL_useful_file_name="${LFBFL_new_file_name}"
197       LFBFL_extension="${LFBFL_useful_file_name##*.}"
198       if [[ LFBFL_i_verbose -eq 1 ]]; then
199         echo "LFBFL_useful_file_name: ${LFBFL_useful_file_name}"
200         echo "LFBFL_extension: ${LFBFL_extension}"
201       fi
202       for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
203         if [[ "${LFBFL_extension}" != "tpl" ]]; then
204           break
205         fi
206         LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
207         LFBFL_extension="${LFBFL_useful_file_name##*.}"
208         if [[ LFBFL_i_verbose -eq 1 ]]; then
209           echo "LFBFL_useful_file_name: ${LFBFL_useful_file_name}"
210           echo "LFBFL_extension: ${LFBFL_extension}"
211         fi
212       done
213       LFBFL_comment_prefix=""
214       if [[ "${LFBFL_extension}" == "sh" ]]; then
215         LFBFL_comment_prefix="# "
216       fi
217       if [[ "${LFBFL_extension}" == "sql" ]]; then
218         LFBFL_comment_prefix="-- "
219       fi
220       if [[ "${LFBFL_extension}" == "tex" ]]; then
221         LFBFL_comment_prefix="% "
222       fi
223       if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
224         LFBFL_comment_prefix="// "
225       fi
226       if [[ LFBFL_i_verbose -eq 1 ]]; then
227         echo "LFBFL_comment_prefix: ${LFBFL_comment_prefix}"
228       fi
229       #-----------------------------------------------------------------
230       # Split if necessary old_file_name, new_file_name, etc. to have
231       # comment lines below max_comment_line_length.
232       LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
233       # -----
```

```
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command1}"\
      "${LFBFL_split_score_command_properties1}"
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
      "${LFBFL_split_score_command_properties2}"
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command1}"\
      "${LFBFL_split_score_command_properties1}"
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
      "${LFBFL_split_score_command_properties2}"
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. -----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
    then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from\n"
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to\n"
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_log_directory_change -eq 1 ]] && [[\
      "${LFBFL_new_file_directory}"\
      != "${LFBFL_old_file_directory}"\
    ]]; then
```

```
280         if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
281         then
282           LFBFL_new_comment+="\n"
283         fi
284         LFBFL_new_comment+="${LFBFL_comment_prefix}"
285         LFBFL_new_comment+="${LFBFL_timestamp}"
286         LFBFL_new_comment+=": This file was moved from\n"
287         LFBFL_new_comment+="${LFBFL_old_file_directory2}"’"\n’
288         LFBFL_new_comment+="${LFBFL_comment_prefix}to\n"
289         LFBFL_new_comment+="${LFBFL_new_file_directory2}"’".’
290       fi
291       if [[ LFBFL_i_verbose -eq 1 ]]; then
292         echo "LFBFL_new_comment: ${LFBFL_new_comment}"
293       fi
294       # Find line with ©Copyright. -------------------------------------
295       LFBFL_copyright_line_number=$(
296         grep -n ’©Copyright’ "${LFBFL_new_file_path}"\
297         | head --lines=1\
298         | cut -d ’:’ -f 1
299       )
300       if [[ LFBFL_i_verbose -eq 1 ]]; then
301         echo "LFBFL_copyright_line_number:"\
302           "${LFBFL_copyright_line_number}"
303       fi
304       if [[ LFBFL_copyright_line_number -eq 0 ]]; then
305         notify-send "${LFBFL_send_summary_2}"\
306           "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
307         if [[ LFBFL_i_verbose -eq 1 ]]; then
308           echo "No copyright line. Skipping file."
309         fi
310         continue
311       fi
312       # Get total number of lines. -------------------------------------
313       LFBFL_line_count=$(
314         wc -l\
315         < "${LFBFL_new_file_path}"
316       )
317       if [[ LFBFL_i_verbose -eq 1 ]]; then
318         echo "LFBFL_line_count: ${LFBFL_line_count}"
319       fi
320       # Compute number of lines after. ---------------------------------
321       LFBFL_lines_after=$((
322         LFBFL_line_count - LFBFL_copyright_line_number
323       ))
324       if [[ LFBFL_i_verbose -eq 1 ]]; then
325         echo "LFBFL_lines_after: ${LFBFL_lines_after}"
```

```
    fi
    # Update file. ----------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_copyright_line_number}"\
        "${LFBFL_new_file_path}"
      echo -e "${LFBFL_new_comment}"
      tail --lines="${LFBFL_lines_after}" "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. ----------------------------------------------
    git add "${LFBFL_new_file_path}"
    LFBFL_renaming_happened=1
  done
  readonly LFBFL_renaming_happened
  if [[ LFBFL_renaming_happened -eq 1 ]]; then
    # Commit ----------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    git commit --message="${LFBFL_message}"
  fi
}
```

2.52 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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```
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#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_library="../DevOrSysAdminScripts/git_helpers/"
LFBFL_library+="git_checks.libr.sh"
# shellcheck source=git_checks.libr.sh
source "${LFBFL_library}"

declare -ig LFBFL_git_hook_error=0
if ! check_files; then
  LFBFL_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_git_hook_error=1
fi
readonly LFBFL_git_hook_error
if [[ LFBFL_git_hook_error -eq 1 ]]; then
  exit 1
fi
exit 0
```

2.53 IMAP_transfert.py

```
"""
This file is part of DevOrSysAdminScripts library.

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DevOrSysAdminScripts is distributed in the hope
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Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don't need "pass" XD.
    """


def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
```

```
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """
```

```
    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print("Creating new folder in to account.")
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
```

```
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
        ),
    )
    parser.add_argument(
        "from_login",
        type=str,
        help="Email account login of user one.",
    )
    parser.add_argument(
        "from_password",
        type=str,
        help="Email account password of user one.",
    )
    parser.add_argument(
        "to_login",
        type=str,
        help="Email account login of user two.",
    )
    parser.add_argument(
        "to_password",
        type=str,
        help="Email account password of user two.",
    )
```

```
    parser.add_argument(
        "--to_new_folder",
        type=str,
        default="",
        help="Transfert emails into a new folder of this name.",
    )
    parser.add_argument(
        "--from_host",
        type=str,
        default="imap.gmail.com",
        help="Host of first email account",
    )
    parser.add_argument(
        "--from_port",
        type=int,
        default=993,
        help="Port on host of first email account",
    )
    parser.add_argument(
        "--to_host",
        type=str,
        default="imap.gmail.com",
        help="Host of second email account",
    )
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
            "-----------------------------------------------------\n"
            "IMAP transfert:\n"
            "-----------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
```

```
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

2.54 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
---------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
```

```
---------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
---------------------------------------------------------------------
```

2.55 miscellaneous/funny.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
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#
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unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  echo "hahaha" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
    fi
  done
  [[ LFBFL_i_result -eq 1 ]] && echo "I had a good laugh."
}

```

```
protected_funny(){
  # Always works regarding the echoes.
  echo "hahaha" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

2.56 miscellaneous/spampty_data.txt

```



a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

2.57 miscellaneous/yet_another_long_long_long_long_long_long_long_long_long_filename_s

2.58 README.md.tpl

```
# DevOrSysAdminScripts

[![CodeFactor-badge]][CodeFactor-package-page]
[![Codacy-badge]][Codacy-package-page]
![GitHub-top-language-badge]
![GitHub-code-size-in-bytes-badge]
[![Security:bandit:badge]][Bandit-GitHub.com]
[![Code-style:black:badge]][Black-GitHub.com]
[![Imports:isort:badge]][Isort-GitHub.io]
[![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
[![Linting:pylint:badge]][Pylint-GitHub.com]
[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

```

```
I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.

But the files "pre-commit" and "post-commit" were not renamed,
since it is not possible to give them other names to use them as
pre-commit/post-commit hooks in `git`.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts

[Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
3dd6ef85e8db4460991d1ebc404b68dd

[Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

[GitHub-top-language-badge]: https://img.shields.io/github/\
languages/top/llyaudet/DevOrSysAdminScripts
```

```

[GitHub-code-size-in-bytes-badge]: https://img.shields.io/github/\
languages/code-size/llyaudet/DevOrSysAdminScripts

[Security:bandit:badge]: https://img.shields.io/badge/\
security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

2.59 unicode.libr.php

```
<?php
/*
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as published by the Free Software Foundation,
either version 3 of the License,
```

```
or (at your option) any later version.

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This file was renamed from "unicode.php" to "unicode.libr.php".
*/

function decimal_code_point_to_UTF8(
  $i_code_point_in_decimal_notation
){
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
  if($i_code_point_in_decimal_notation < 128){
    return chr($i_code_point_in_decimal_notation);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 256 * (2**3)){
    $i_first_byte_base_value = 192;
    $i_last_byte_significant_bits = (
      $i_code_point_in_decimal_notation % (2**6)
    );
    $i_first_byte_significant_bits = intdiv(
      (
        $i_code_point_in_decimal_notation
        - $i_last_byte_significant_bits
      ),
      2**6,
    );
    $s_result = chr(
      $i_first_byte_base_value
      + $i_first_byte_significant_bits
    );
    $s_result .= chr(
      $i_continuation_base_value
```

```
56        + $i_last_byte_significant_bits
57      );
58      return $s_result;
59    }
60    // 1110xxxx 10xxxxxx 10xxxxxx
61    if($i_code_point_in_decimal_notation < 256 * 256){
62      $i_first_byte_base_value = 224;
63      $arr_arr_data_per_byte_reverse = [
64        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
65        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
66        ["i_bits" => 4, "i_base_value" => $i_first_byte_base_value],
67      ];
68      $i_rest = $i_code_point_in_decimal_notation;
69      $s_result = "";
70      foreach($arr_arr_data_per_byte_reverse as $arr_data){
71        // _Pragma unroll
72        // J'aurais pu "roller" les 4 cas des ifs.
73        // Mais pour les perfs, il faudrait tout unroller vu le nombre
74        // d'itérations. Et j'aimerais bien comparer du code unrollé
75        // aux 2 niveaux à la main avec celui unrollé par le compilateur
76        // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
77        // l'intérieur d'abord (pour ne pas refaire le même boulot)
78        // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
79        // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
80        $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
81        $i_rest = intdiv(
82          $i_rest - $i_significant_bits,
83          2**$arr_data["i_bits"],
84        );
85        $s_result = (  // .= or =.
86          chr($i_significant_bits + $arr_data["i_base_value"])
87          .$s_result
88        );
89      }
90      return $s_result;
91    }
92    // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
93    if($i_code_point_in_decimal_notation < 256 * 256 * (2**5)){
94      $i_first_byte_base_value = 240;
95      $arr_arr_data_per_byte_reverse = [
96        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
97        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
98        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
99        ["i_bits" => 3, "i_base_value" => $i_first_byte_base_value],
100     ];
101     $i_rest = $i_code_point_in_decimal_notation;
```

```
102     $s_result = "";
103     foreach($arr_arr_data_per_byte_reverse as $arr_data){
104       // _Pragma unroll
105       $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
106       $i_rest = intdiv(
107         $i_rest - $i_significant_bits,
108         2**$arr_data["i_bits"],
109       );
110       $s_result = (  // .= or =.
111         chr($i_significant_bits + $arr_data["i_base_value"])
112         .$s_result
113       );
114     }
115     return $s_result;
116   }
117   throw new Exception(
118     "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
119     ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
120     ." 3 octets au lieu de 4 :face-palm:."
121     ." Du coup, ça peut donner quelques \"surprises\"."
122     ." Sinon, y'a aussi ASCII dans MySQL."
123     ." Et comme ça joue sur la mémoire et les performances,"
124     ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
125     ." :)"
126   );
127 }
128
129
130
131 function hexa_code_point_to_UTF8(
132   $s_code_point_in_hexadecimal_notation
133 ){
134   $s_code_point_in_hexadecimal_notation = strtoupper(
135     $s_code_point_in_hexadecimal_notation
136   );
137   $i_code_point_in_decimal_notation = 0;
138   $i_ord_0 = ord('0');
139   $i_ord_A = ord('A');
140   $i_digit_offset = -$i_ord_0;
141   $i_letter_offset = 10 - $i_ord_A;
142   for(
143     $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
144     $i < $i_max;
145     ++$i
146   ){
147     $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
```

```
148       $i_code_point_in_decimal_notation *= 16;
149       if(ctype_digit($s_hexa_digit)){
150         $i_code_point_in_decimal_notation += (
151           ord($s_hexa_digit) + $i_digit_offset
152         );
153         continue;
154       }
155       if(ctype_xdigit($s_hexa_digit)){
156         $i_code_point_in_decimal_notation += (
157           ord($s_hexa_digit) + $i_letter_offset
158         );
159         continue;
160       }
161       throw new Exception("Not an hexadecimal digit.");
162     }
163     return decimal_code_point_to_UTF8(
164       $i_code_point_in_decimal_notation
165     );
166   }
167
168
169
170   function check_string_is_valid_ASCII($s_string){
171     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
172       if(ord($s_string[$i]) > 127){
173         throw new Exception(
174           "Non-ASCII character found at octet position "
175           .$i
176           ., value "
177           .ord($s_string[$i])
178           ."."
179         );
180       }
181     }
182     return true;
183   }
184
185
186
187   function check_file_is_valid_ASCII($s_file_name){
188     $s_string = file_get_contents($s_file_name);
189     if($s_string === false){
190       throw new Exception("File ".$s_file_name." not found.");
191     }
192     return check_string_is_valid_ASCII($s_string);
193   }
```

```
194
195
196
197 function check_string_is_valid_UTF8($s_string){
198   $i_continuation_octet_needed = 0;
199   $i_character_start_position = 0;
200   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
201     $i_current_octet = ord($s_string[$i]);
202     if($i_continuation_octet_needed > 0){
203       if($i_current_octet < 128 || $i_current_octet >= 192){
204         throw new Exception(
205           "Non-UTF8 character found at start position "
206           .$i_character_start_position
207           ., octet at position "
208           .$i
209           ." has value "
210           .$i_current_octet
211           ." which is not a continuation octet."
212         );
213       }
214       --$i_continuation_octet_needed;
215       continue;
216     }
217
218     $i_character_start_position = $i;
219     if($i_current_octet < 128){ // 0xxxxxxx ASCII
220       continue;
221     }
222     if($i_current_octet >= 128 && $i_current_octet < 192){
223       throw new Exception(
224         "Non-UTF8 character found at start position "
225         .$i_character_start_position
226         ., octet at position "
227         .$i
228         ." has value "
229         .$i_current_octet
230         ." which is a continuation octet."
231       );
232     }
233     if($i_current_octet >= 192 && $i_current_octet < 224){
234       // 110xxxxx 10xxxxxx
235       $i_continuation_octet_needed = 1;
236       continue;
237     }
238     if($i_current_octet >= 224 && $i_current_octet < 240){
239       // 1110xxxx 10xxxxxx 10xxxxxx
```

```
240         $i_continuation_octet_needed = 2;
241         continue;
242       }
243       if($i_current_octet >= 240 && $i_current_octet < 248){
244         // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
245         $i_continuation_octet_needed = 3;
246         continue;
247       }
248       throw new Exception(
249         "Non-UTF8 character found at start position "
250         .$i_character_start_position
251         ., octet at position "
252         .$i
253         ." has value "
254         .$i_current_octet
255         ." which is invalid."
256       );
257     }
258     return true;
259   }
260
261
262
263   function check_file_is_valid_UTF8($s_file_name){
264     $s_string = file_get_contents($s_file_name);
265     if($s_string === false){
266       throw new Exception("File ".$s_file_name." not found.");
267     }
268     return check_string_is_valid_UTF8($s_string);
269   }
270
271
272
273   /*
274   ?>
275   <?php
276   require_once("./unicode.libr.php");
277   var_dump(hexa_code_point_to_UTF8("002B"));
278   var_dump(hexa_code_point_to_UTF8("00E6"));
279   var_dump(hexa_code_point_to_UTF8("1400"));
280   var_dump(hexa_code_point_to_UTF8("10111"));
281   // https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
282   // Some invisible unicode characters that are a problem:
283   // 2000-200F 2028-202F 205F 2060-206F
284   $arr = [
285     '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
```

```
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
foreach($arr as $s_hexa){
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  | sed -e 's/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
}
//*/

```

2.60 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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#
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# that it will be useful,
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#
```

```
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# the GNU General Public License
# along with DevOrSysAdminScripts.
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#
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# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
  fi
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed_point_build* ]]; then
    LFBFL_fixed_point_build="--fixed_point_build"
  fi
  readonly LFBFL_verbose
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  ./build_and_checks.exec.sh "." "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"
}

upgrade_build_and_checks "$@"
```

2.61 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1 file_path
  # $2 download_URL
  # $3 correct_sha512
  # $4 optional --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 wget_sha512 $*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  if [[ ! -f "$1" ]];
  then
    wget "${LFBFL_verbose}" -O "$1" "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum "$1"\
    | cut -f1 -d' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]];
  then
    echo "$1 does not have correct sha512"
    echo "wanted $3"
    echo "found ${LFBFL_present_sha512}"
    exit
  fi
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !